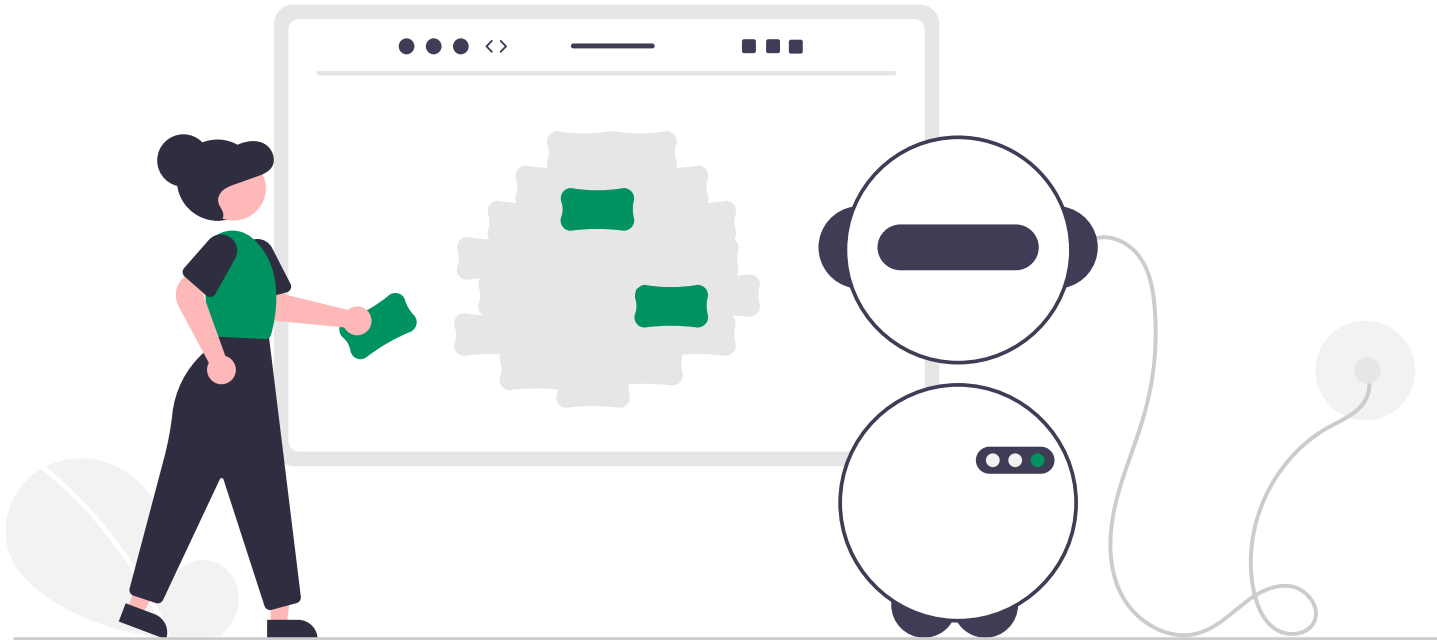




Co-creative UIs & Interaction with Generative Systems

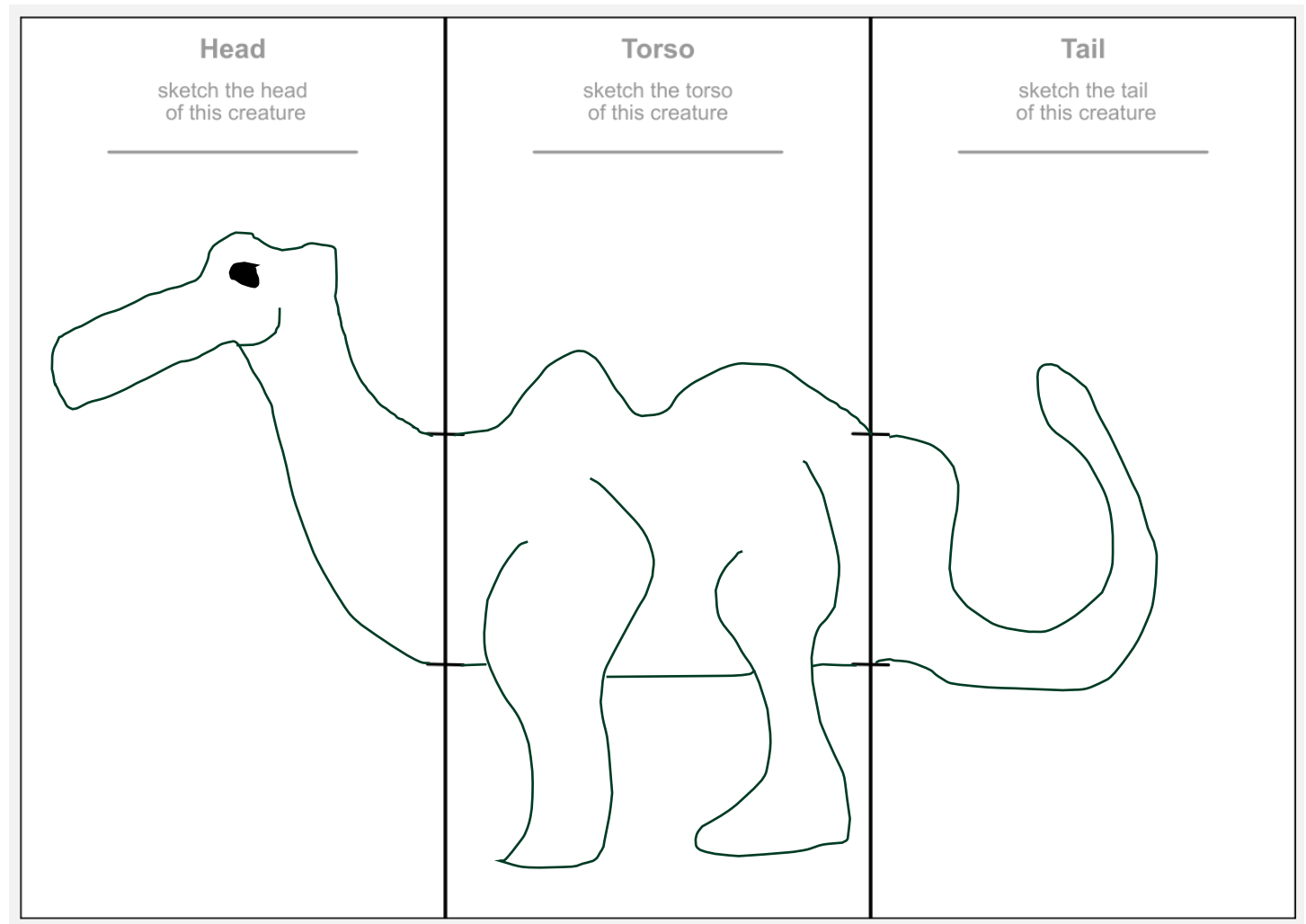
Create a creature!

You'll need:

- A pen
- This sheet 
- Two more people
(everyone can be part of multiple "teams" here)

Draw the first part of your creature (connect to the small lines), then fold the drawn part behind the sheet and pass on the sheet to someone else.

Continue drawing without looking at the previous parts!



Recap & Last Quiz

Lecture Quiz

Name and briefly describe three technical tasks / components required for interaction with a voice assistant.

Voice Recognition / Natural Language Understanding (NLU)

This component converts spoken language into text and interprets the user's intent.

Example: Understanding that "What's the weather tomorrow?" is a weather forecast request.

Dialogue Model / Dialogue Management

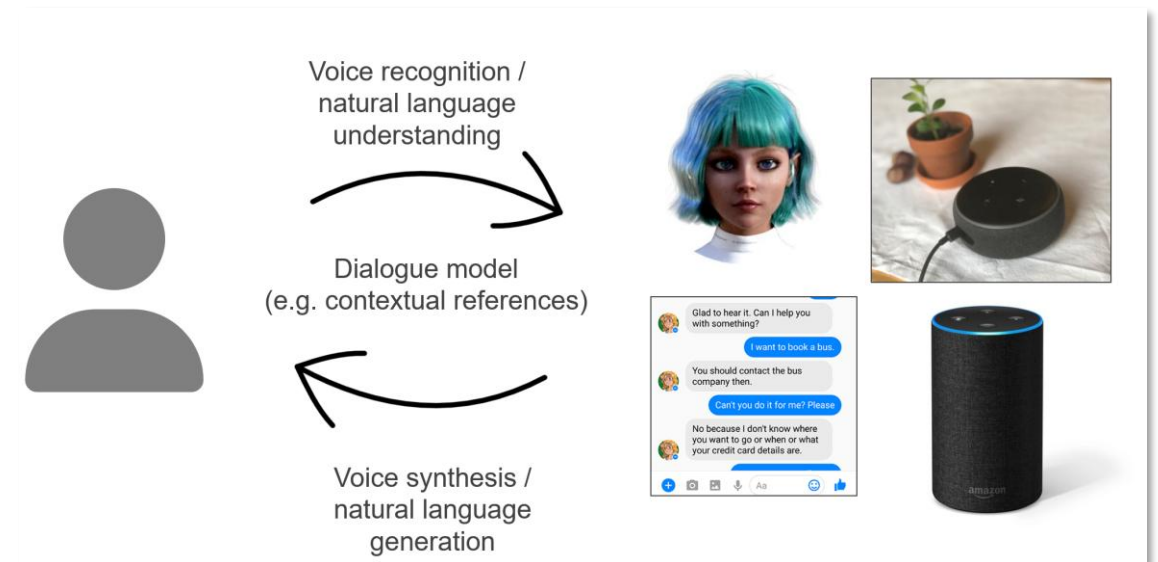
This manages the flow and context of the conversation. It keeps track of previous interactions and handles contextual references or follow-up questions.

Example: If the user asks "What about Friday?", the system understands it refers to the weather forecast mentioned before.

Voice Synthesis / Natural Language Generation (NLG)

This component generates a response and converts it into spoken audio.

Example: The assistant replies with synthesized speech such as "Tomorrow will be sunny with 22 degrees."



Lecture Quiz

Describe **benefits** and **drawbacks** of allowing free text input vs giving fixed input options in chatbots.

Free text input is **flexible and expressive**, the user can ask anything in any way.

But it puts **more burden on the system to understand intent**, and **users can get lost not knowing what to say**.

Fixed input options are **more predictable** and **easier** to use, especially for first-time users, but they **limit what the user can express** and can feel restrictive for complex tasks.

Free text input is **more conversational** because of how open-ended it is.

[Free text input] might give [users] the **illusion that the system can solve anything**.

Free text: [...] harder to **control** [for developers]

Lecture Quiz

Describe at least three possibilities for **influencing the user's perception of the 'personality'** of a voice assistant.

Voice characteristics: Different tone, pitch, speed, or accent.

Response behavior: Polite greetings, jokes, empathy, or level of detail in answers.

Language style: Formal, friendly, humorous, or professional wording.

Visual appearance, how the avatar or device presents itself.

Lecture Quiz

You're asked to **improve a photo editing software** (e.g. Photoshop) by **integrating a voice assistant** as a new feature. Describe ideas of how the voice assistant might help the user in this context (optional: feel free to add a UI sketch), and what special (technical) capabilities this assistant would require.

Assistant for learning how to use the software and/or finding features, e.g.:

"the agent would listen to your request and respond either with an LLM - based voice reply, or by showing things on your screen with a bright dot, a "laser pointer" similar to the mouse laser pointer in PowerPoint. The agent would **need access to the program structure** so it knows where to look for which feature."

Capabilities needed: Integration with editing APIs, understanding of visual elements, contextual awareness of selected areas, and ability to map natural language to specific tool actions."

Assistant for faster execution:

"When the user knows what they want, they can **just say it instead of searching menus**, e.g. "increase saturation by 20" or "make it brighter". Useful because the hands and eyes are busy on the image."

Referencing across speech and direct manipulation:

"The assistant should **track the mouse pointer**."

GUI integration for monitoring:

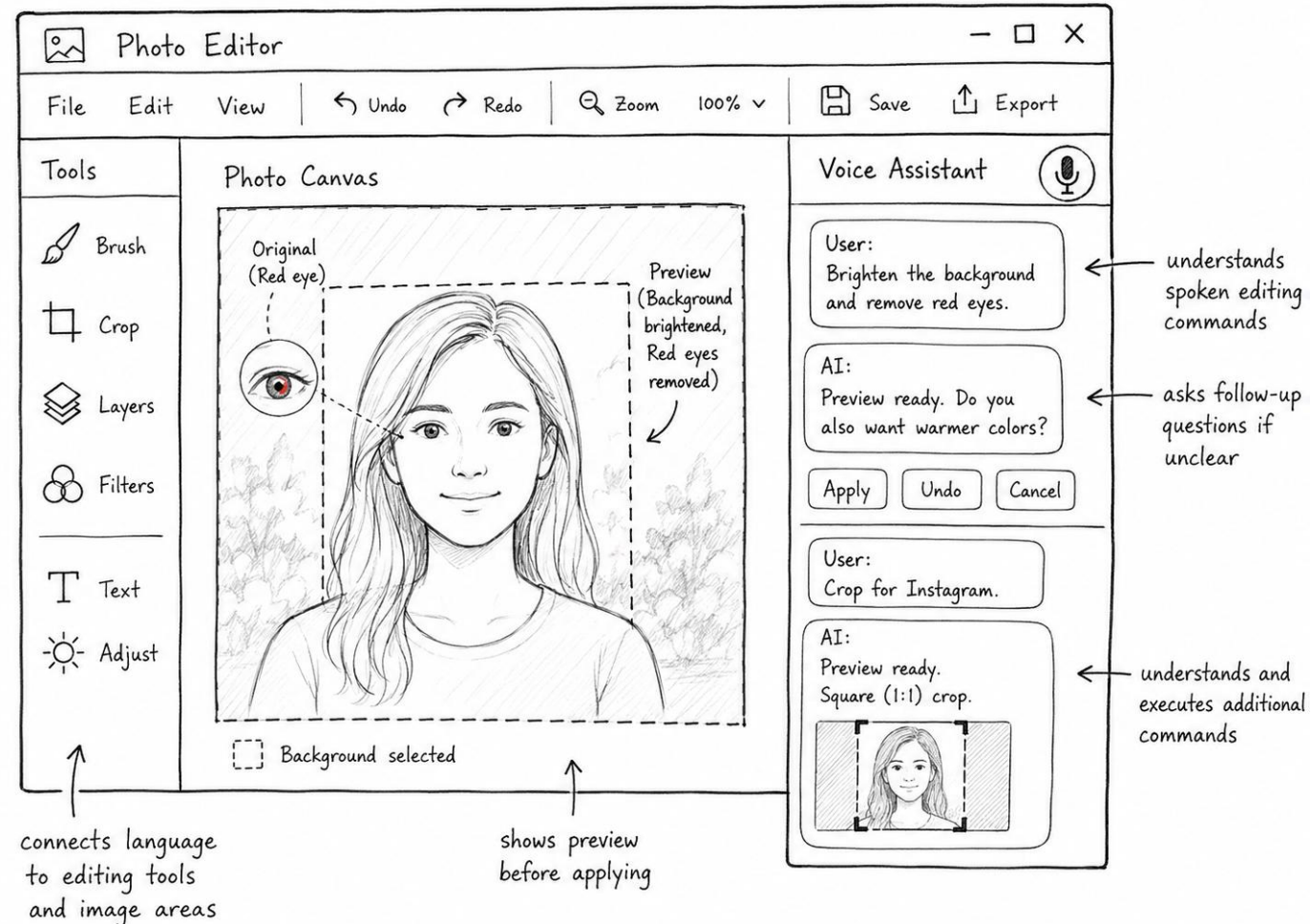
"A useful UI addition would be a small floating **voice indicator** showing what the system heard and what action it is about to take, so the user stays in control."

Lecture Quiz

Integrating voice, chat UI, and image (pre)view

For example, the user could say: “make the background brighter”, “remove the red eyes”, “crop the image for Instagram”, or “make the face look more natural”. The assistant could then **apply the change directly or show a preview first**. This would be useful because photo editing programs often have many tools and menus, so voice input could **reduce searching and clicking**.

It could also work as a mixed initiative system. The user gives the goal, and the **assistant helps with a possible action list**. For example, if the user says “improve this photo”, the assistant could ask: “Do you want better lighting, sharper details, or color correction?” This is helpful because the user’s intention may not always be clear.



This lecture

Learning goals

After this session, you should be able to:

- ✓ **Describe** typical components/aspects of a generative system and how they might be considered when designing a UI for such a system
- ✓ **Analyse** a UI / interaction design
 - ✓ By **applying** the discussed twelve principles of mixed-initiative interaction
 - ✓ By **applying** the discussed “COFI” design dimensions
 - ✓ By **applying** the discussed analysis framework for (turn-based) co-creation
- ✓ **Explain a broader perspective** on interaction with generative systems (e.g. when designing such systems and/or as a user of related products), by thinking beyond the currently dominant "system-as-a-content-generator" role distribution



Activity

Your creative use of computers

Which **digital artefacts** do you personally create regularly?

What's something you work on / create / do at a computing device that **feels** creative to you?



Intro

Terminology
Examples

“Creative”

Clarification of terminology

- Can machines be creative?
- This lecture:
 - Create ~ generate structured data
 - No intention to assign thought, artistic value, authorship, etc. to a machine

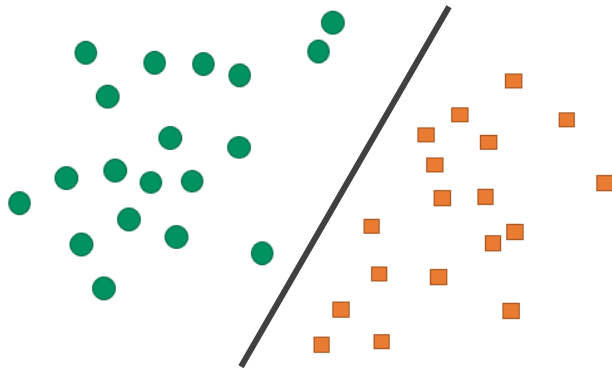
But these are interesting to examine! For example, how does authorship depend on the interaction design when using AI-generated material? As experienced by users? Legally? Philosophically?

„Generative“

Clarification of terminology

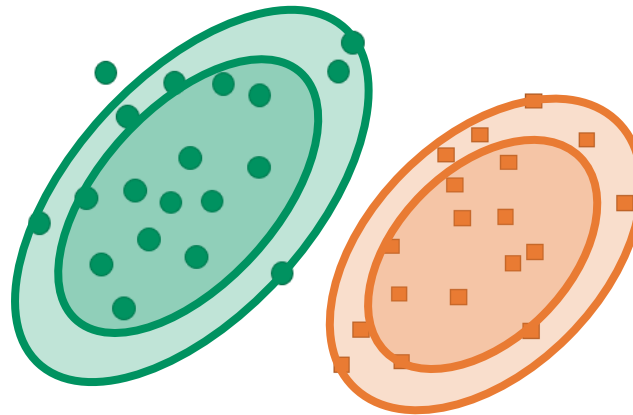
- Here: „Generative system“ = system capable of **generating structured output** (e.g. text, images, game levels, ...)
- Generally in ML: Discriminative vs generative models

Discriminative model learns $p(y|x)$,
e.g. predict $y=cat$ or dog , given $x=pixels$
→ decisions, but cannot generate data



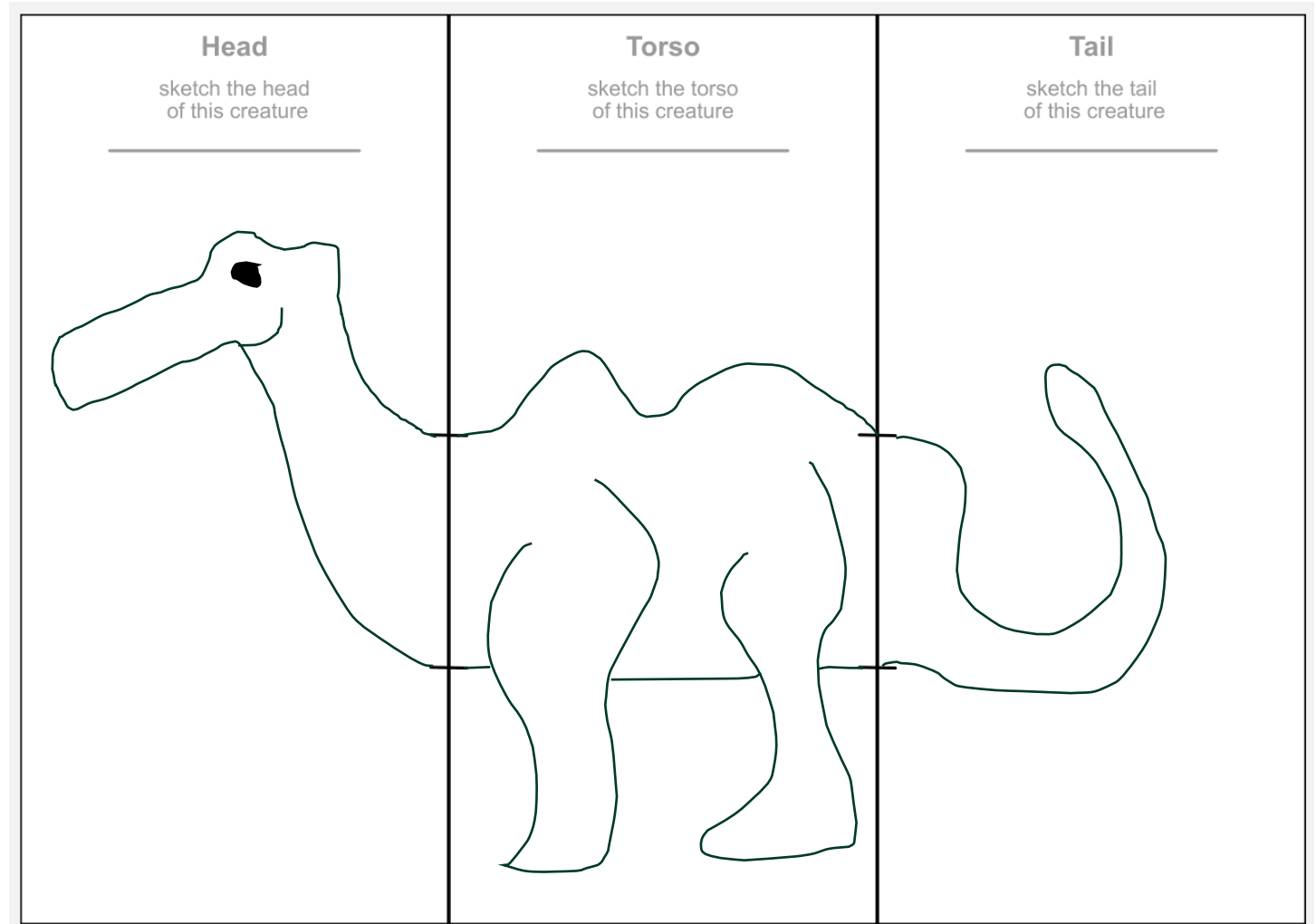
See SVM in lecture 2

Generative model learns $p(y,x)$,
e.g. distribution of cat and dog images
→ can generate new images



Create a creature!

- Complete your creatures now
- Congratulations – you’ve just engaged in human-human co-creation!
- You’ve also realized a specific generative model:
 - If you didn’t look, your creature’s head, torso, and tail were drawn from independent “distributions” (= your ideas on the spot)
 - If you cheated and peeked, you instead implemented a conditional distribution, e.g. $p(\text{tail} \mid \text{head}, \text{torso})$
(read: “distribution of tail, given head and torso”)
- We come back to these concepts in our session on text entry (e.g. auto-completion)



An analog example

Of an interactive generative system - this helps us see the “typical parts”:

Training data:

real species put into the system
(=book)

Assumed data-generating process:

A species has 3 independent parts
head, torso, tail

(in our creature-activity, if you’ve looked at the other parts, you would have realized a conditional process, since the existing parts might then have influenced your part)

Chosen model:

One discrete distribution per part
Uniform (e.g. all heads equally likely)



Generation:

Sample a head, a torso, a tail
Combine



User interface: 3 flaps

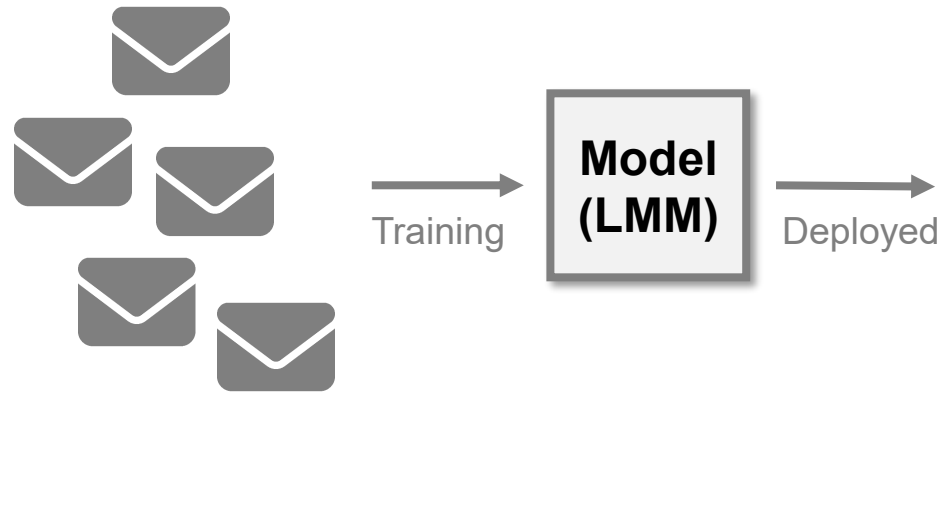


Interactions: Select or flip at random

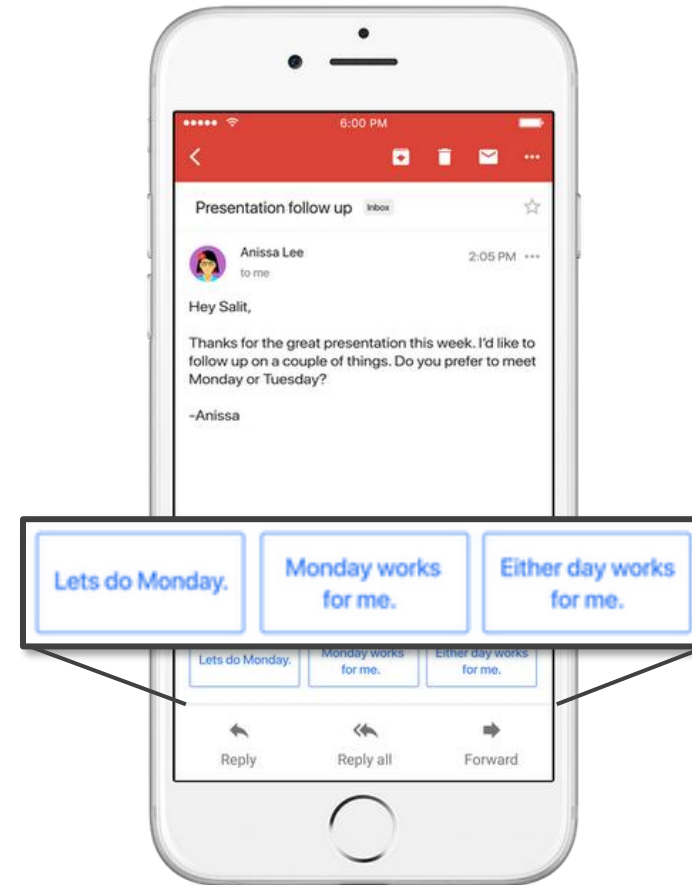
Examples we've seen before

Text suggestions in GMail

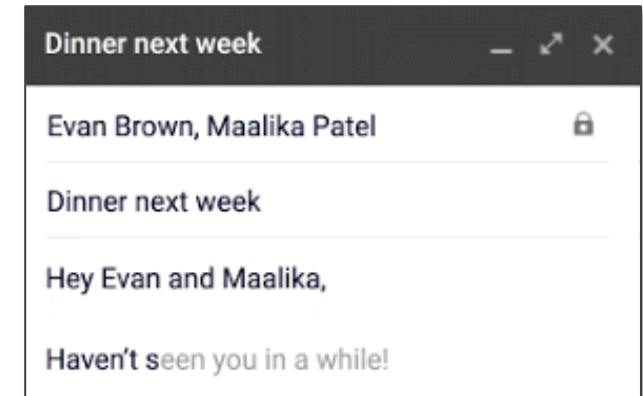
Lots of emails / text



„Smart Reply“



„Smart Compose“



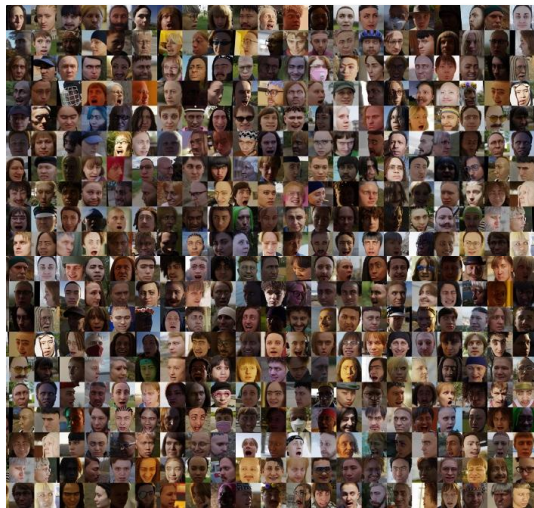
+ lots of AI features for text applications recently...

<https://blog.google/products/gmail/save-time-with-smart-reply-in-gmail/>
<https://ai.googleblog.com/2018/05/smart-compose-using-neural-networks-to.html>

Examples we've seen before

„Smart Portrait Filters“ in Adobe's Photoshop

Portrait dataset

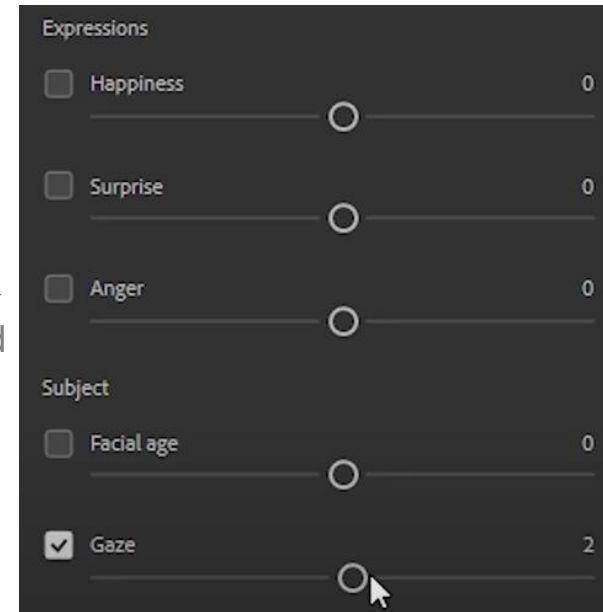


Training

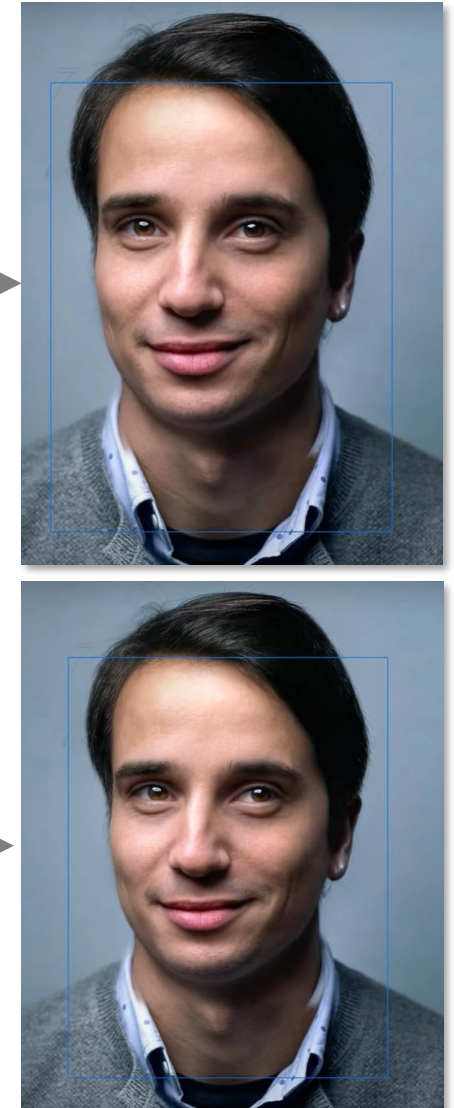
**Model
(NN)**

Deployed

User Interface



User
changes
gaze
direction
with
slider



<https://microsoft.github.io/DigiFace1M/>, <https://blog.adobe.com/en/2020/10/20/photoshop-the-worlds-most-advanced-ai-application-for-creatives.html> <https://blogs.nvidia.com/blog/2020/10/20/adobe-max-ai/>, <https://github.com/NVLabs/stylegan2>

Examples we've seen before Luminate

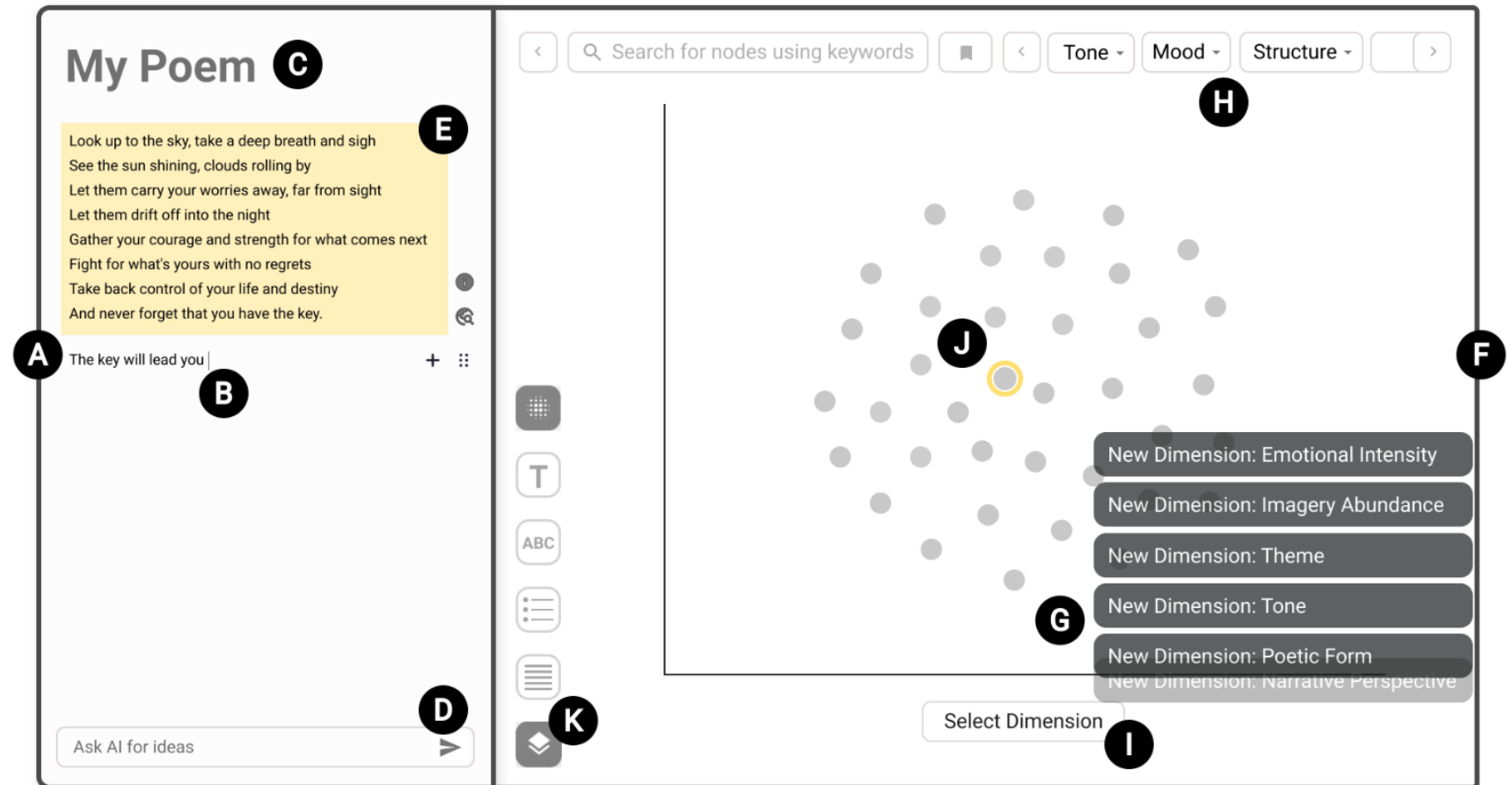


Figure 3: Luminate interface consists of two main sections: (A) text editor and (F) exploration view. Users can (B) type and use various (C) text styles (e.g., TITLE, normal text). They have the option to (D) input a prompt (e.g., "Give me a poem about ocean") to LLM and view (E) one of the generated responses in the text editor. (F) In the exploration view, users can observe (J) a cluster of responses and (H) dimensions. At first, no dimension is selected, to invite users to explore the design space. Users can add dimensions to the axes (I) and arrange responses by their dimension values (see Fig. 5). For optimal experience, we promptly display (G) dimensions as soon as they are generated, allowing users to examine them while responses are being generated. Users can click (K) semantic level icons to adjust the zoom scale and view responses at different levels of detail (see Fig. 8).

[Suh et al., 2024]

Further example

Misty: UI Prototyping Through Interactive Conceptual Blending

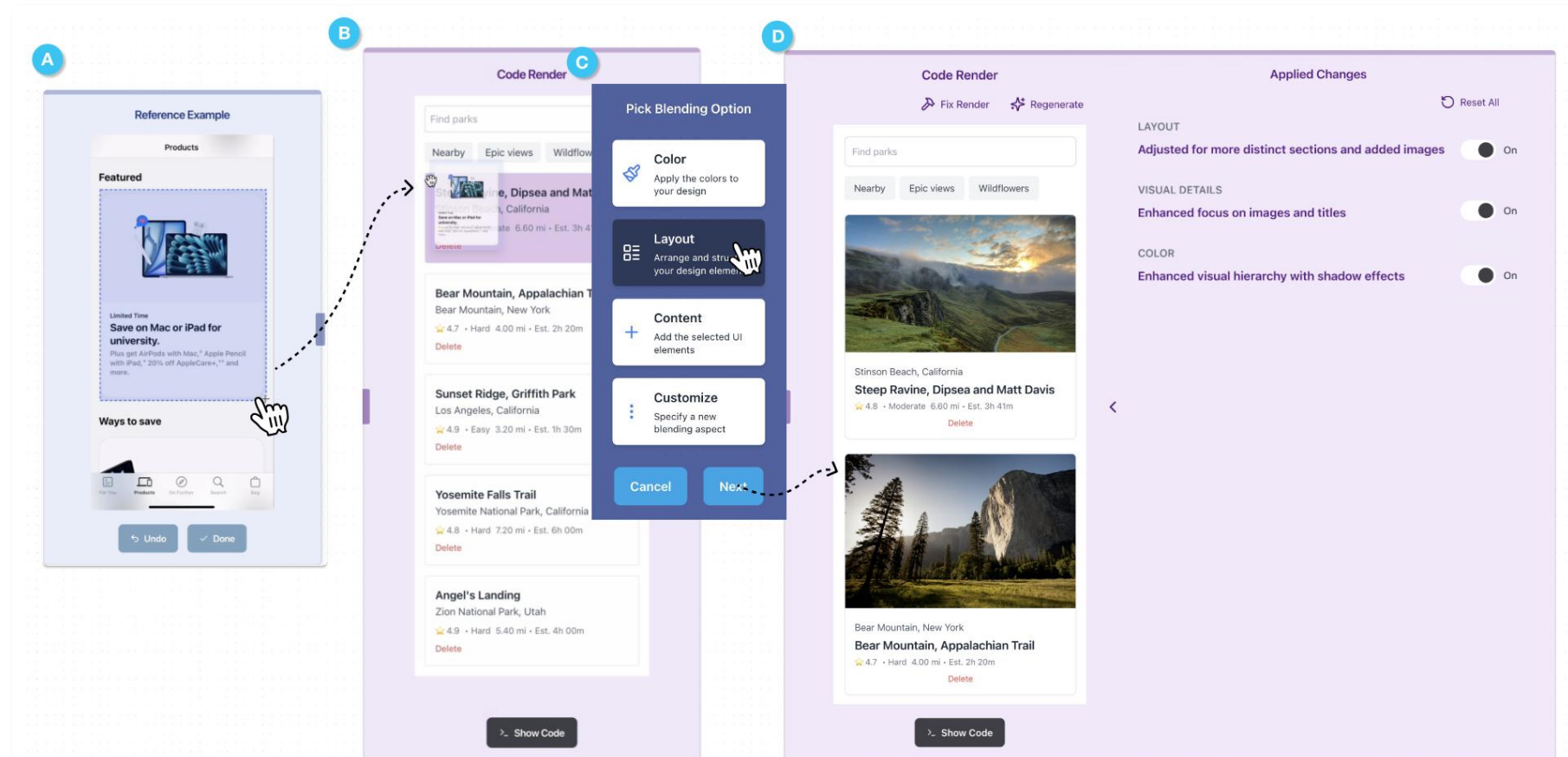


Figure 1: The MISTY system. MISTY embodies the interactive conceptual blending interaction for UI prototyping, where UI developers *blend* selected sections of an example image **A** into work-in-progress UI code **B**. Users can *specify* aspect(s) of the image to blend in **C**. After viewing the blending output **D**, the user can further tweak details by toggling on/off within the *semantic diff* (that is, semantically categorized code changes) between the updated and original UI versions. Initially, the work-in-progress UI shows basic cards with some park information. The user selects the card layout from the example UI, drags it onto their current design's cards, and chooses to blend the "Layout". MISTY then generates updated UI code, preserving the original content but presenting it in a new card layout with images and hierarchical text descriptions, mirroring the example's layout.

[Lu et al., 2025]



3X

Mixed-initiative interaction

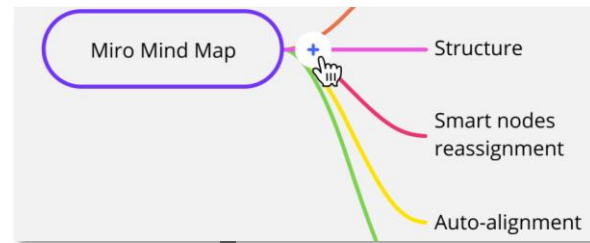
When *both* user and system can do sth in the UI / to the artefact;
originally: a mix of tool-like and agent-like aspects

„Intelligent“ software for creating digital artefacts

Four concepts & perspectives

Creativity support tools

- Support human
- E.g. a mind mapping tool



Co-creative AI

- Joint human-AI work
- E.g. „Duet Draw“



[Oh et al., 2018]

Human
creates

mixed initiative
interaction

System
creates

AI tools

- Add new capabilities
- E.g. „smart portrait filters“



Note: use of the terms varies, e.g. „creativity support tools“ is sometimes an overarching term, which includes co-creative AI etc.

Automation

- Replace human
- E.g. news article generation

Microsoft is cutting dozens of MSN news production workers and replacing them with artificial intelligence

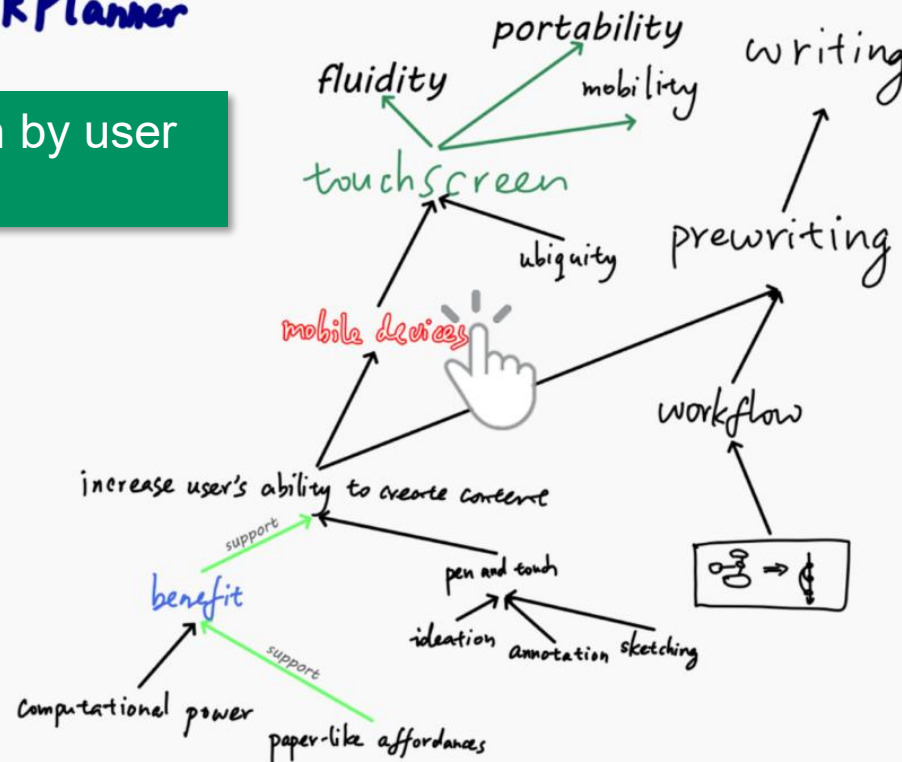
May 29, 2020 at 4:58 pm | Updated May 29, 2020 at 8:43 pm

These concepts are often mixed in UI design

Example: „InkPlanner“ – mind map + AI suggestions

InkPlanner

Mind map written by user
(pen + tablet)

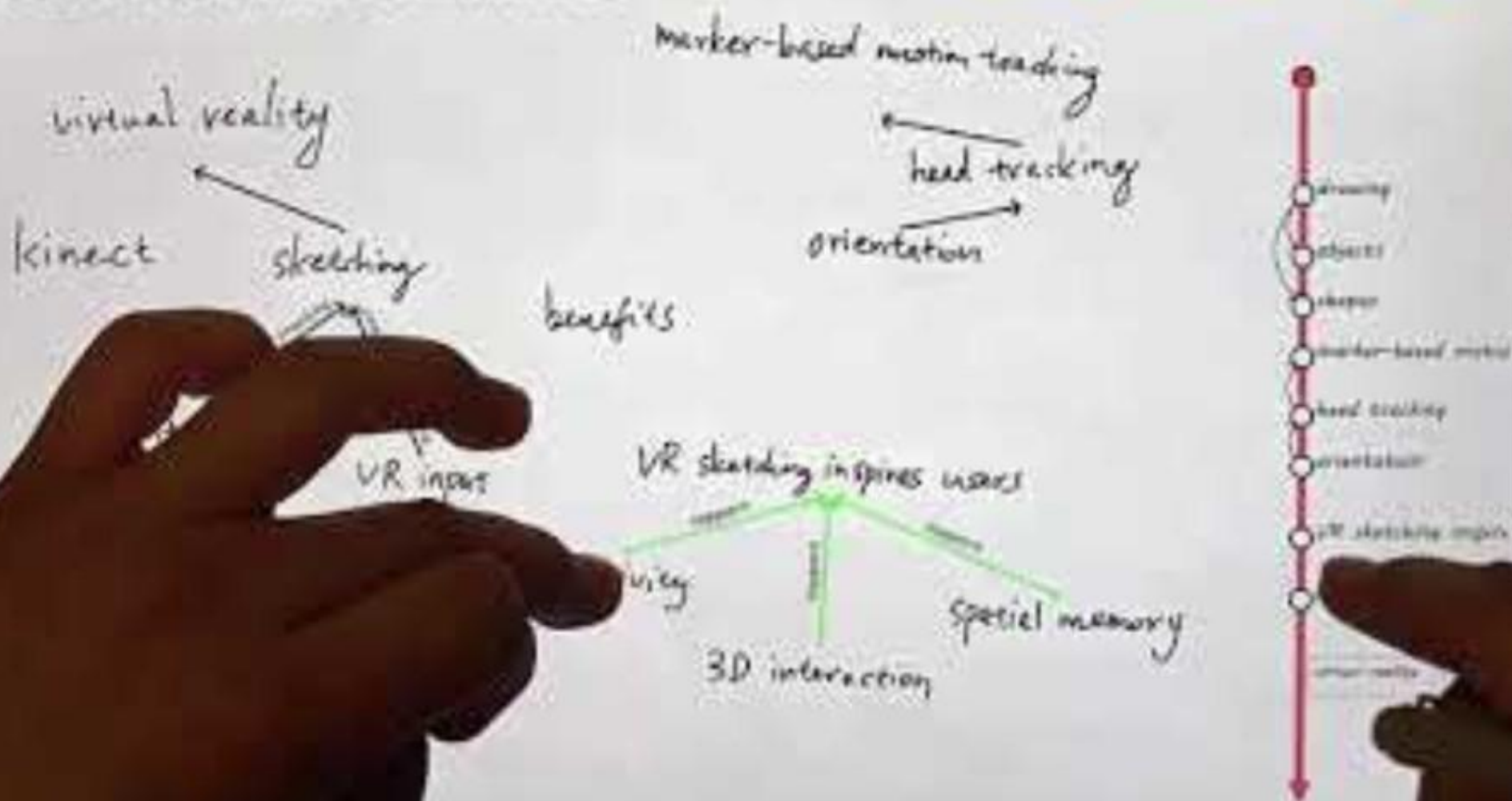


Related words suggested
by the system

[Lu et al., 2019]

Video (suggestion feature shown at ca. 3:10):
<https://www.youtube.com/watch?v=oVsSkYm3mlk>

NarrativeLine



Mixed-initiative UIs

Twelve long-standing principles

“Recent debate [in 1999] has centered on the relative promise of focusing user-interface research on

- developing new metaphors and **tools that enhance users’ abilities to directly manipulate objects**
- versus directing effort toward developing **interface agents that provide automation.**

In this paper, we review principles that show promise for allowing engineers to **enhance human-computer interaction through an elegant coupling of automated services with direct manipulation.**”

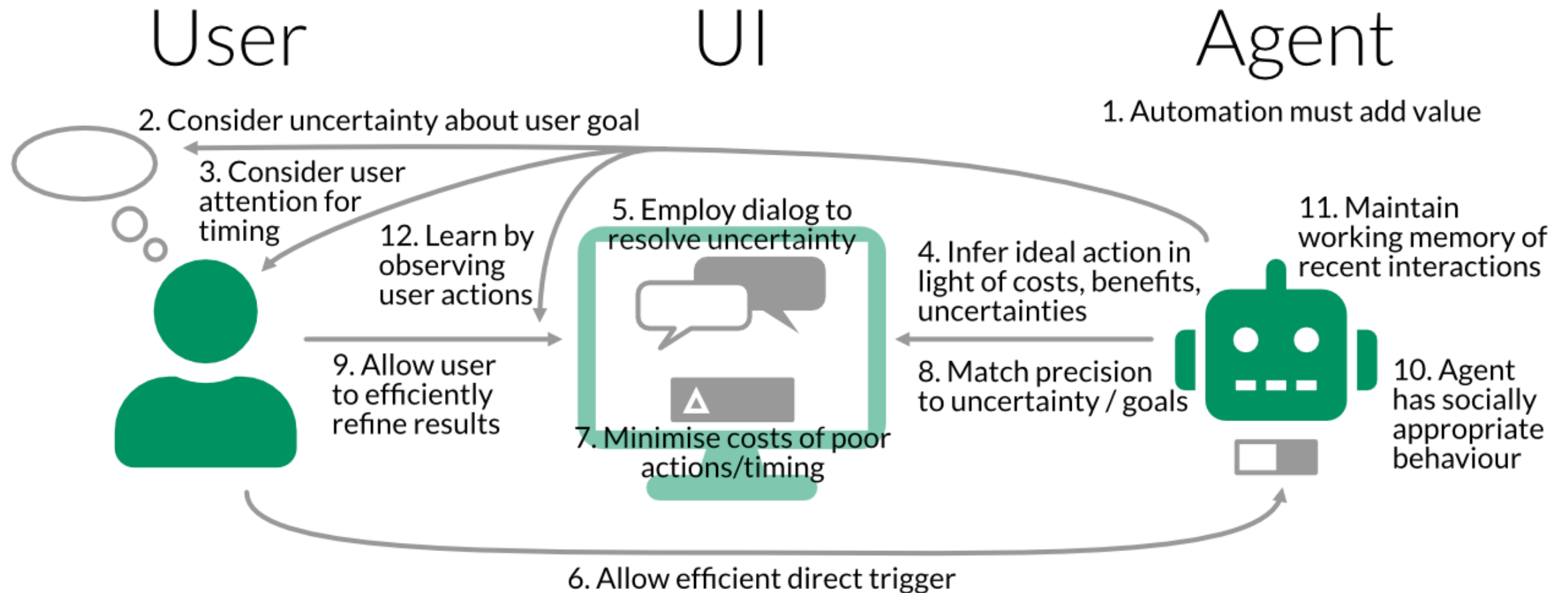
[Horvitz, 1999]



Eric Horvitz,
CSO, Microsoft

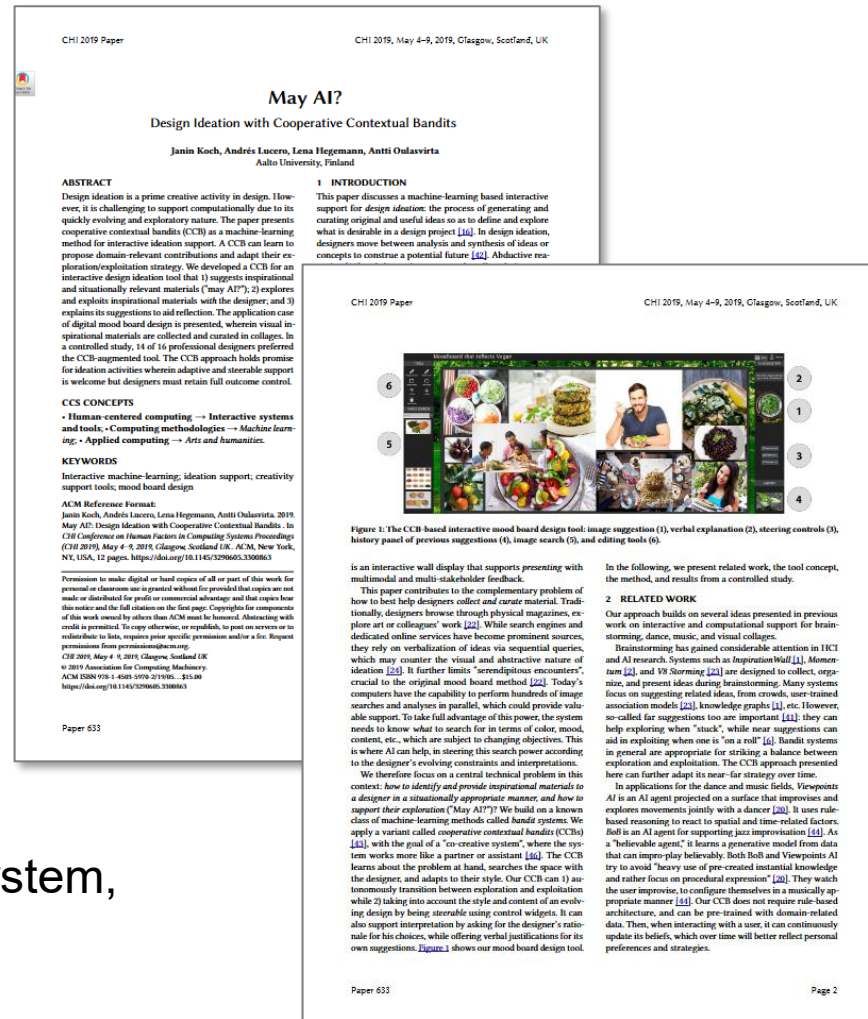
Twelve principles

Illustrated



Case study

We analyse a mixed-initiative co-creative system, using the twelve principles



Janin Koch
Assist. Prof.,
Inria Lille

[Koch et al., 2019]

Case study: Mixed-initiative mood boards

User and system collect images to co-create a mood board

Background
about this use case:

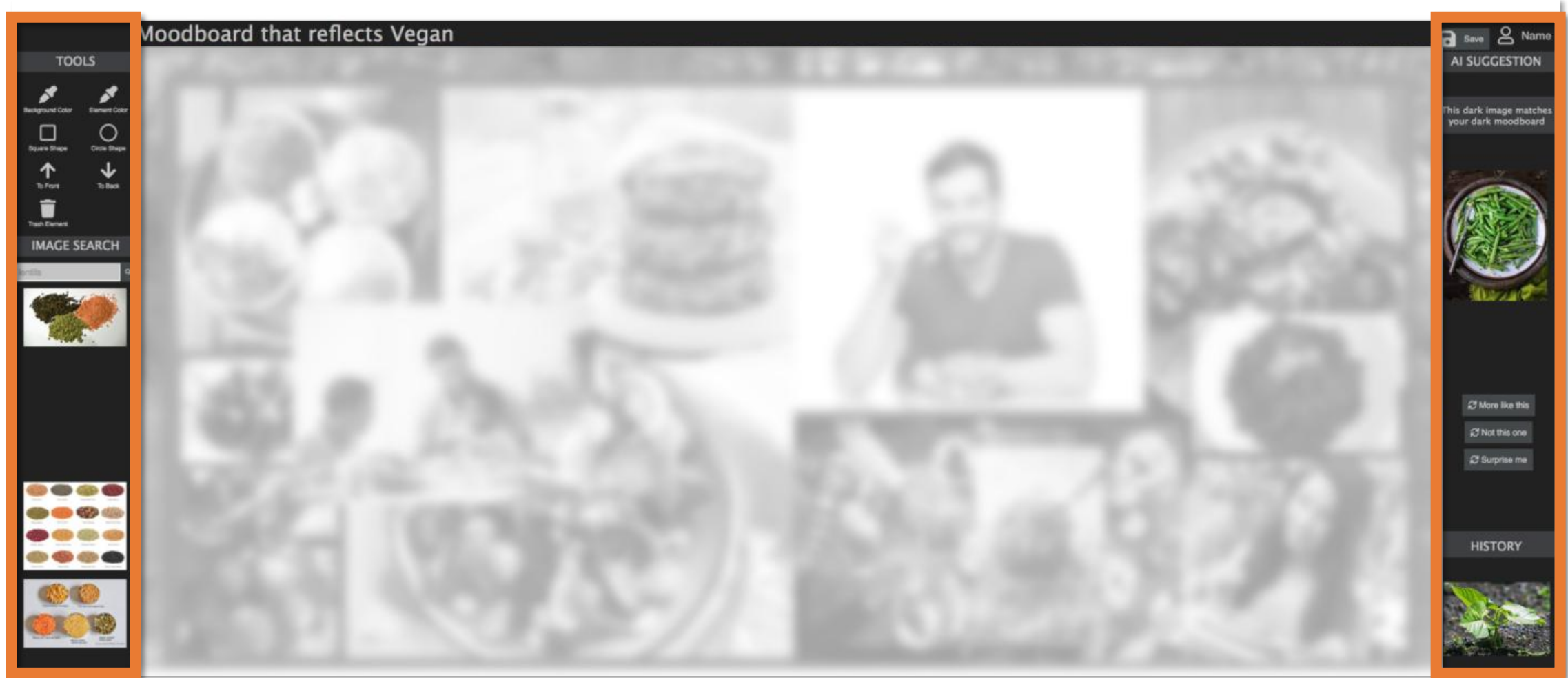
- **Mood board:** A collage to visualise hard-to-express ideas for inspiration and decisions
- **Example:** This could be a moodboard for a startup or brand identity around „vegan eating“



[Koch et al., 2019]

Case study: Mixed-initiative mood boards

User interface



[Koch et al., 2019]

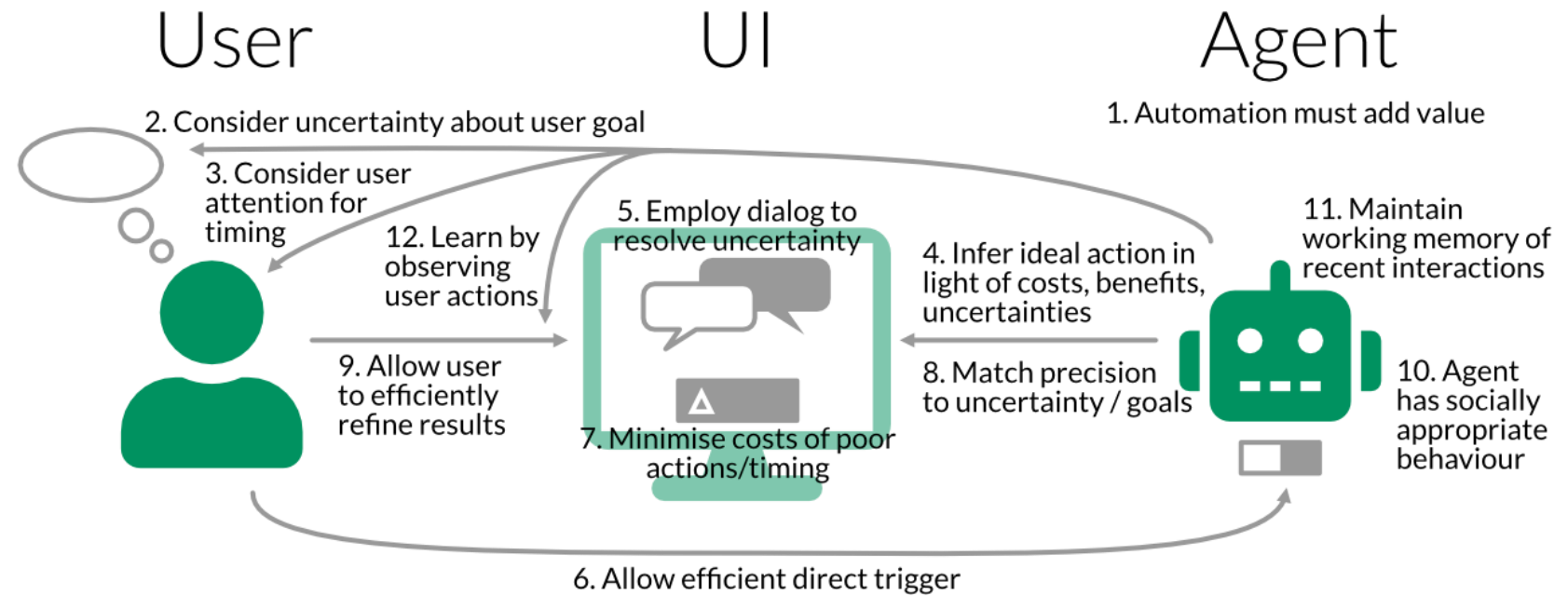
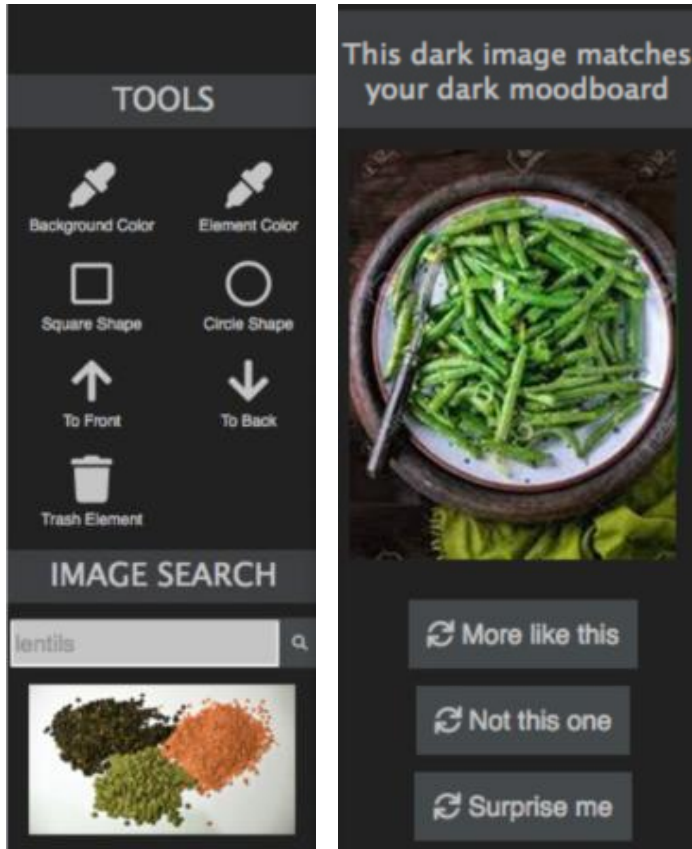
Activity

Identify some of the 12 principles in the moodboard system



Activity

Identify some of the 12 principles in the mood board system

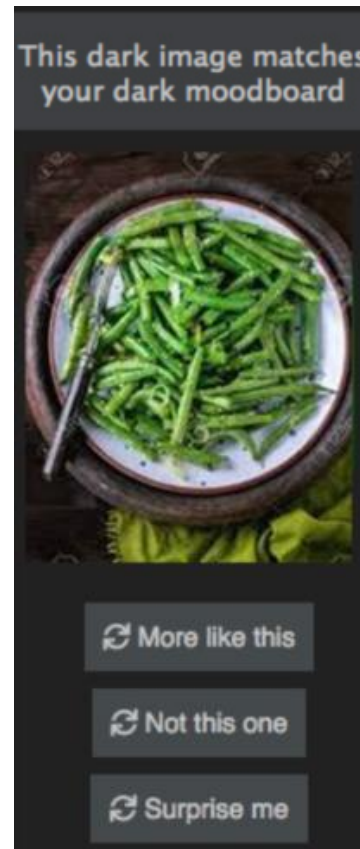
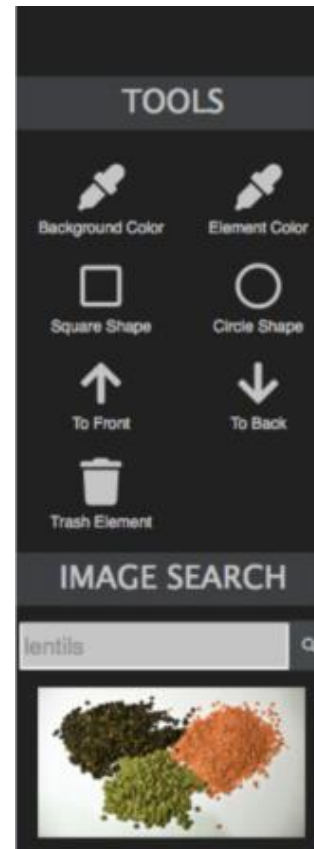


Analysis of the main UI elements and their “roles”

In the mood board system

User's tools

for direct manipulation
(e.g. search and add image)



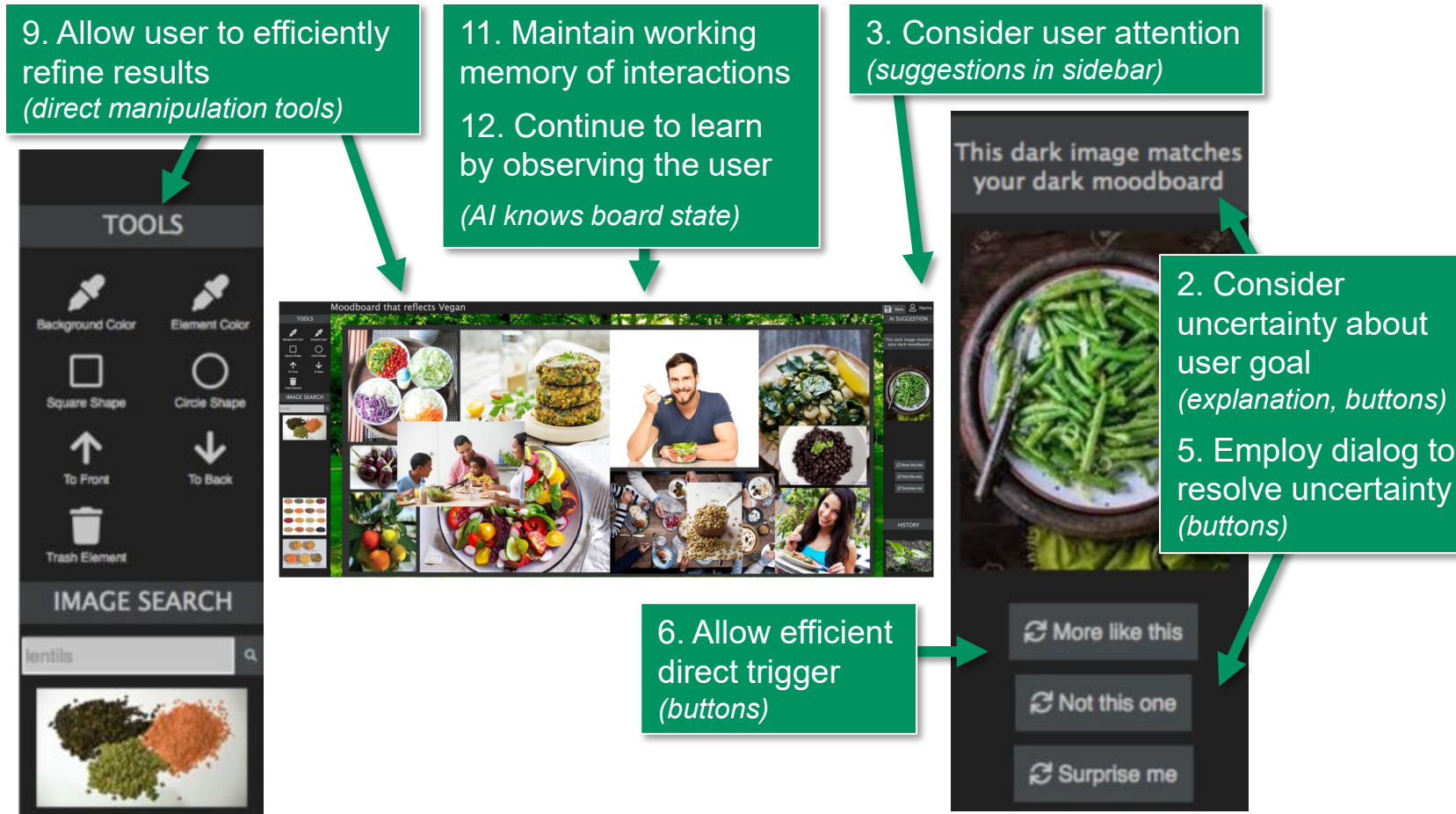
AI suggestions
+ explanations

Buttons
for **feedback**
to the AI

Analysis of the twelve principles

In the mood board system

Note: This is intended as an example but not to say that this is the only “correct” analysis; you might argue for different principles of for these principles in different ways.



Exploration vs exploitation

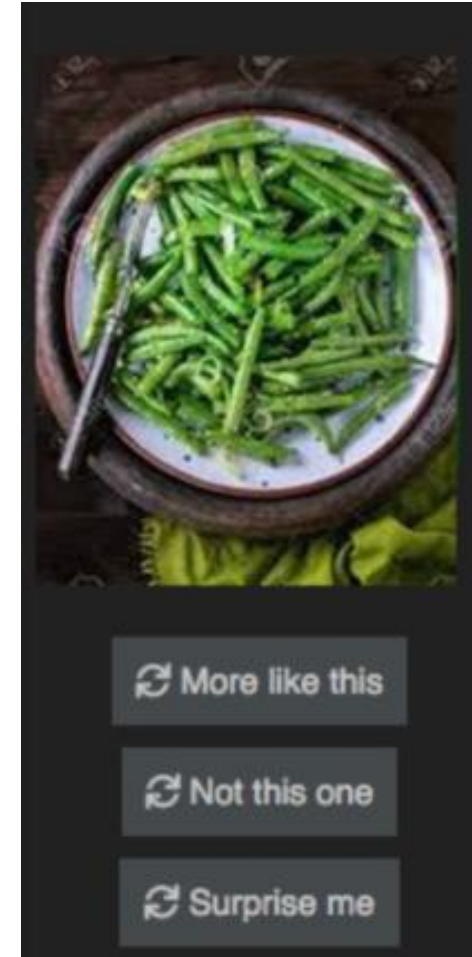
A common tradeoff for co-creative / mixed-initiative systems

- **Exploration**

- AI/system moves away from the current solution
- E.g. buttons „Not this one“, „Surprise me“

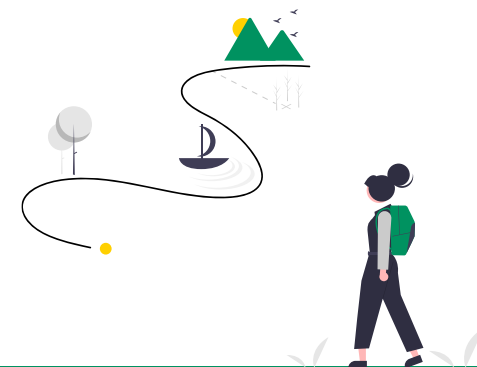
- **Exploitation**

- AI/system iterates on the current solution
- E.g. button „More like this“



Note: Remember session 3 on rec. systems?

We covered exploration vs exploitation there, too. Here, it is about creation, not consumption: Informing or inspiring the user's own next "creative step" (e.g. what they will next do on their mood board)!



Co-creative interaction

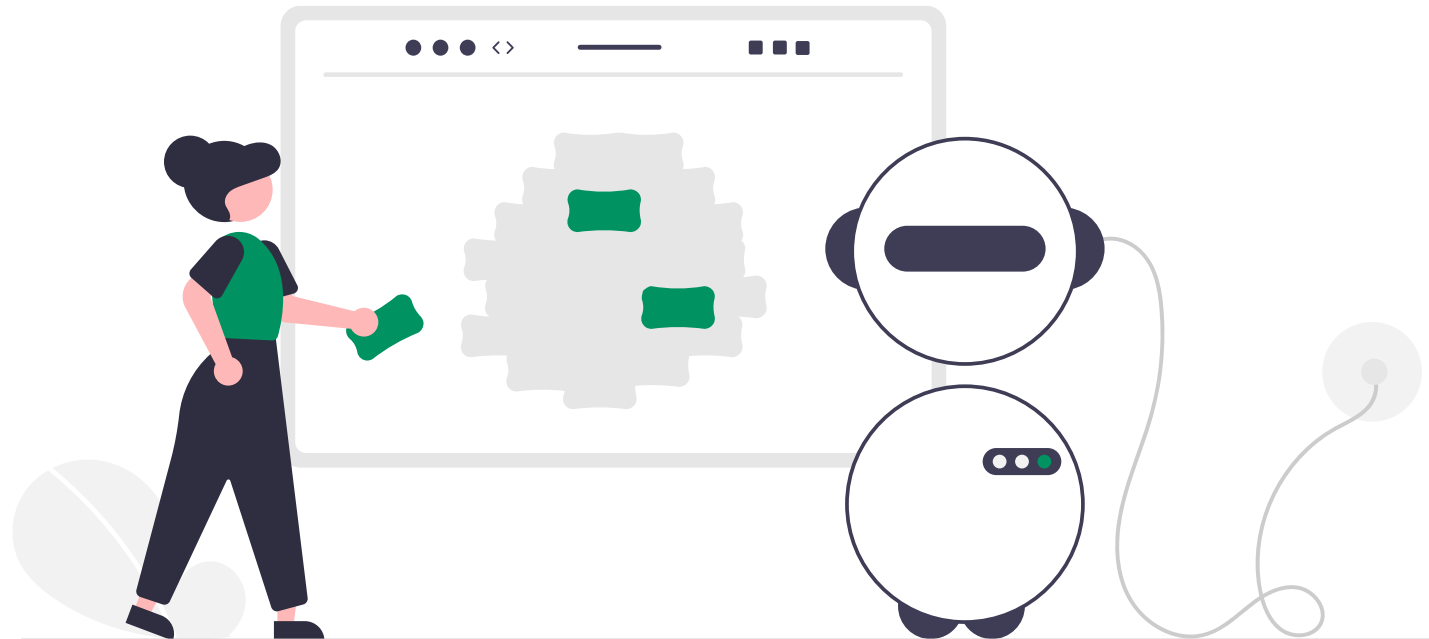
Both human and AI contribute towards creating a digital artefact, considering what the other has done so far and perhaps „communicating“ about the shared creation / process

Co-creative interaction

Key idea:

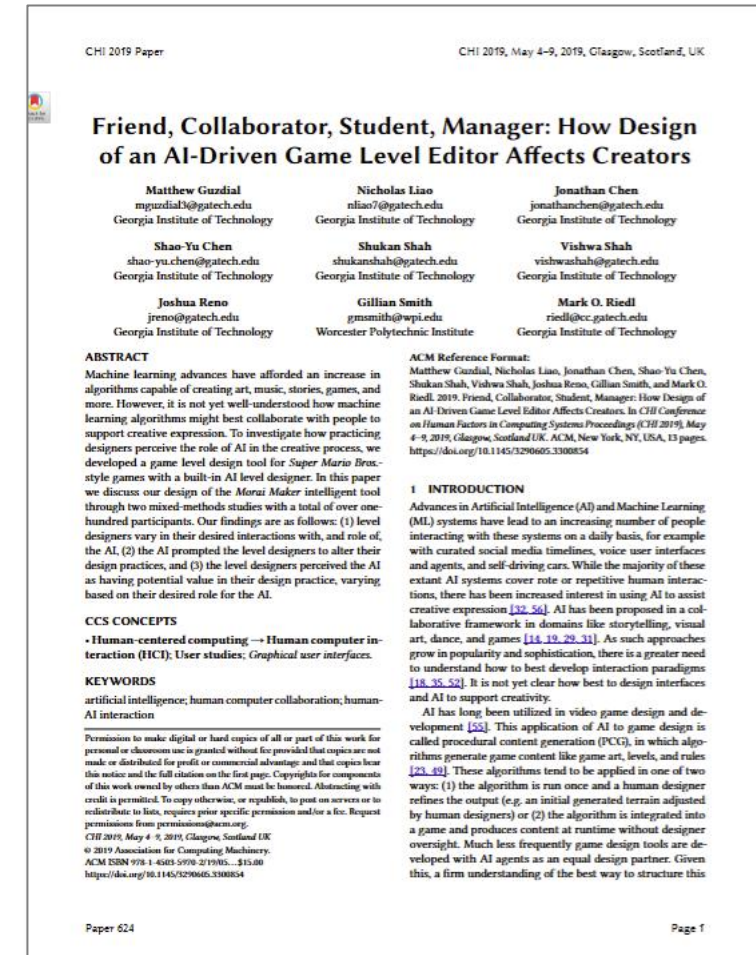
Both human and AI contribute to creating a result;
typically a digital artefact that requires a sequence
of creative decisions

(as in the mood board example)



Case study: Co-creative level design

Level editor with an AI „partner“

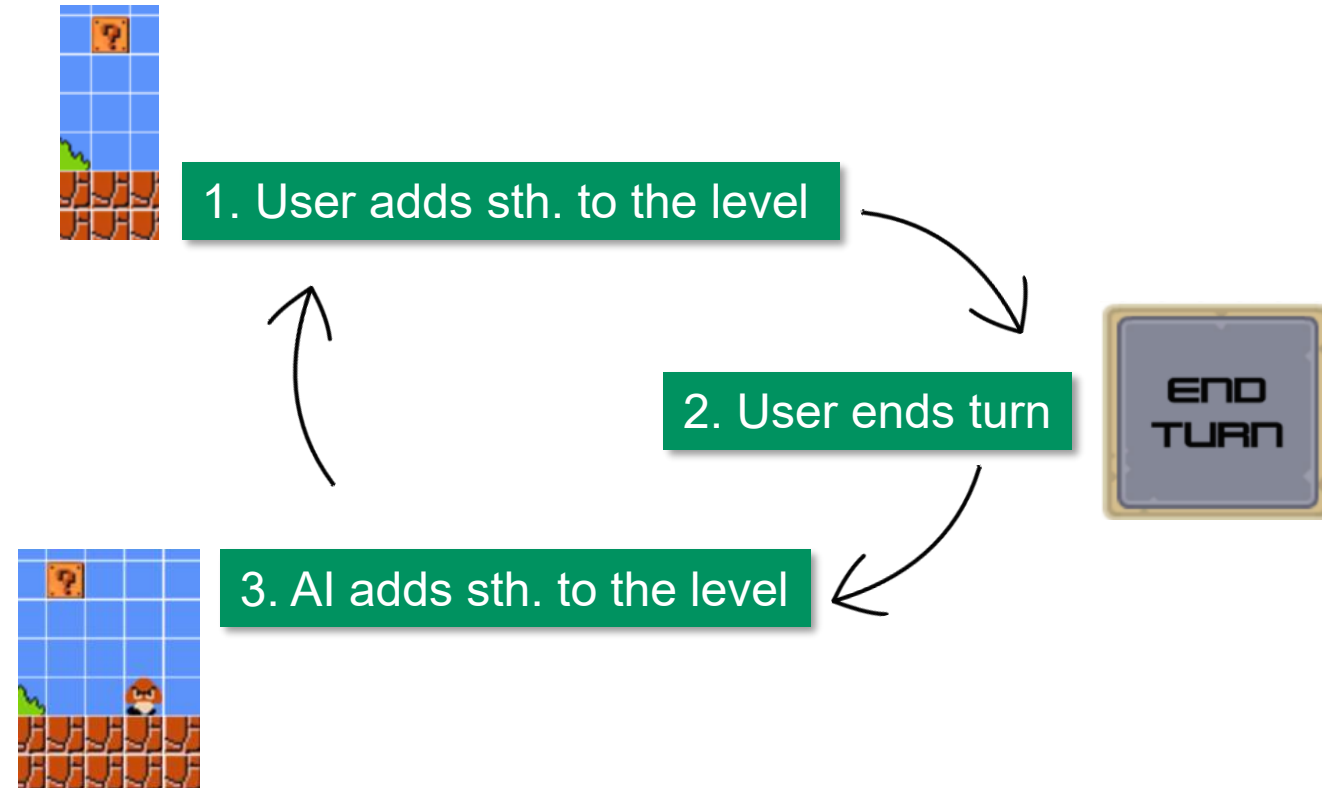
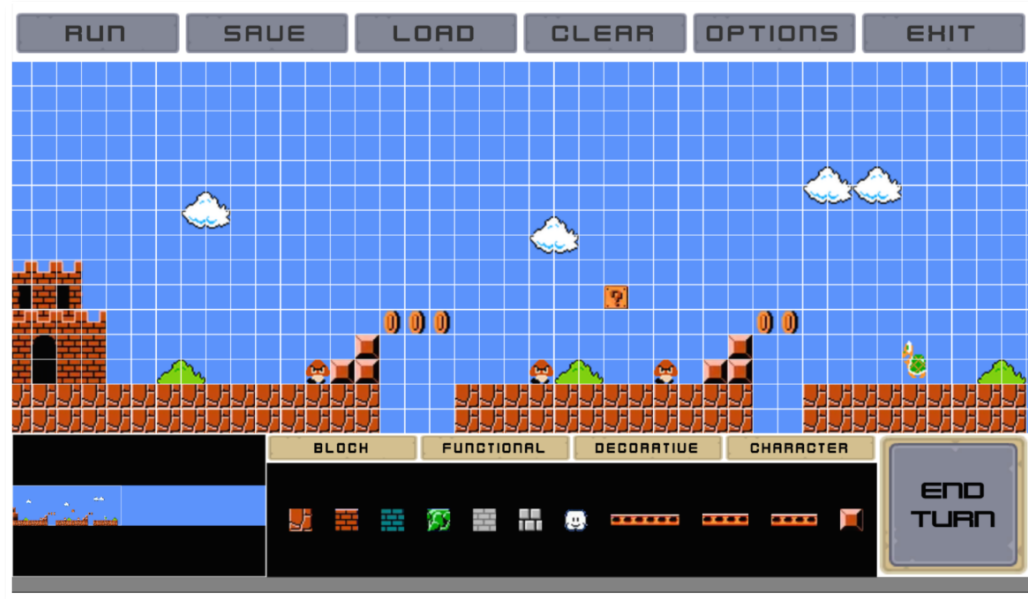


[Guzdial et al., 2019]

<https://www.youtube.com/watch?v=kvsnmswMkTk>

Case study: Co-creative level design

Interaction design for co-creation as “taking turns”



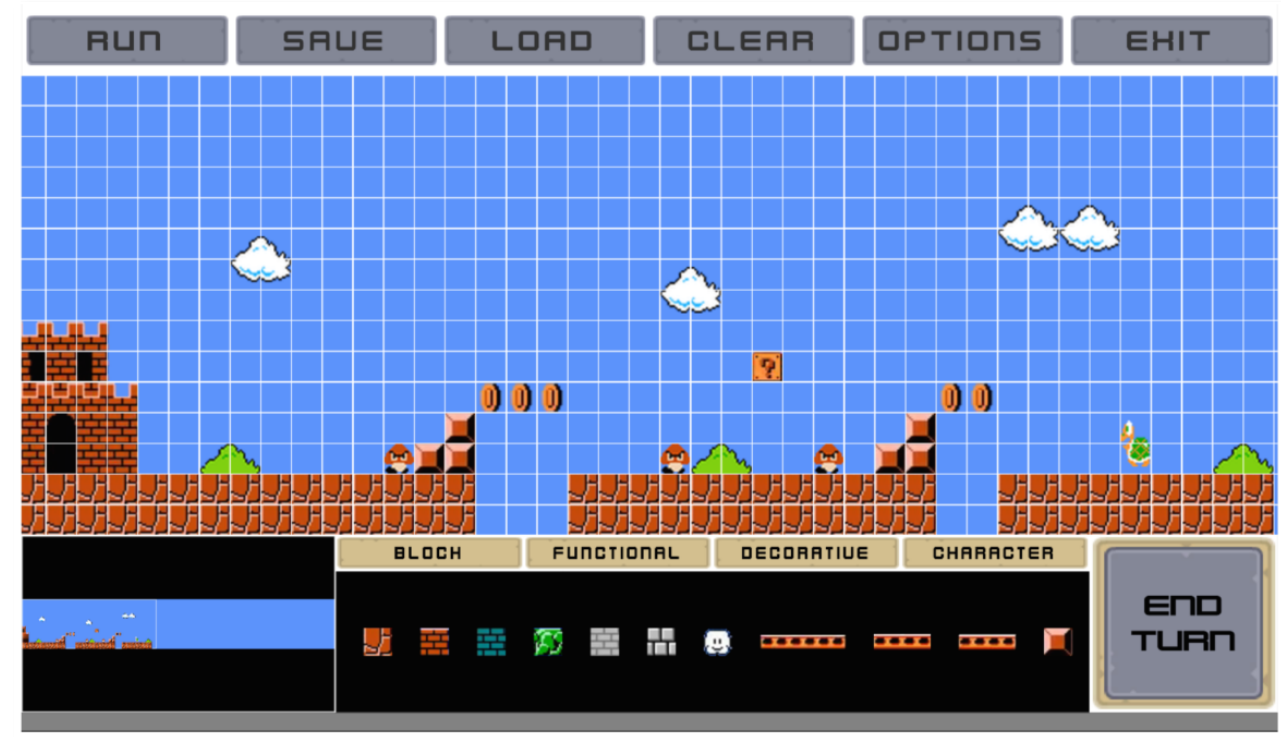
[Guzdial et al., 2019]

Case study: Co-creative level design

Some findings from their user study

*“[...] users varied widely in their expectations and expected **role** of the AI agent”*

Short discussion with your neighbour(s):
How would *you* describe the role of the AI agent here?



[Guzdial et al., 2019]

Case study: Co-creative level design

Identified “roles” from their user study

- **Friend:** Interaction with AI as a fun activity

“Let’s see what my friend comes up with”

- **Collaborator:** AI as an equal design partner

“What I expect from a design partner is one contributing complete(ish) ideas rather than small edits”

- **Student:** AI follows instructions

“It tried to copy things I did and it wasn’t ‘frustrating’ to work with”

- **Manager:** AI gives instructions or judges the user’s design

“[...] I’ll follow along with what it suggested”

[Guzdial et al., 2019]

“Roles” beyond this case study

A lot of work on „AI roles“, for example:

- Hoffman, 2016
 - **Divergent agent:** Generates proposals, finds problems
 - **Convergent agent:** Evaluates proposals
- Kantosalo and Toivonen, 2016
 - **Creation: Generator, Evaluator, Concept definer**
 - **Control (of initiative): Pleasing agent, Provoking agent**
- Negrete-Yankelevich and Morales-Zaragoza, 2014
 - **Generator, Apprentice, Master**



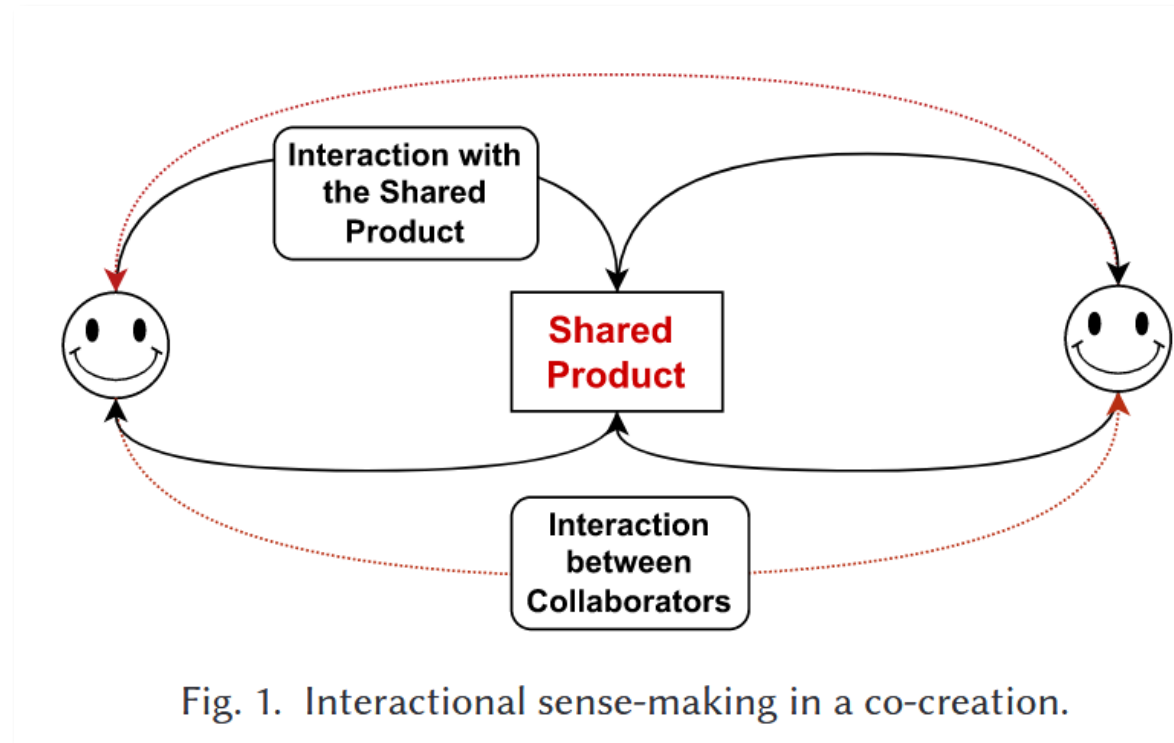
Overview see: Kantosalo & Jordanous, 2020 + Rezwana & Maher, 2022

Structuring co-creation

In a more systematic way
than the various “role”-names

Structuring co-creation

Two ways of interacting with the other “partner” in co-creation



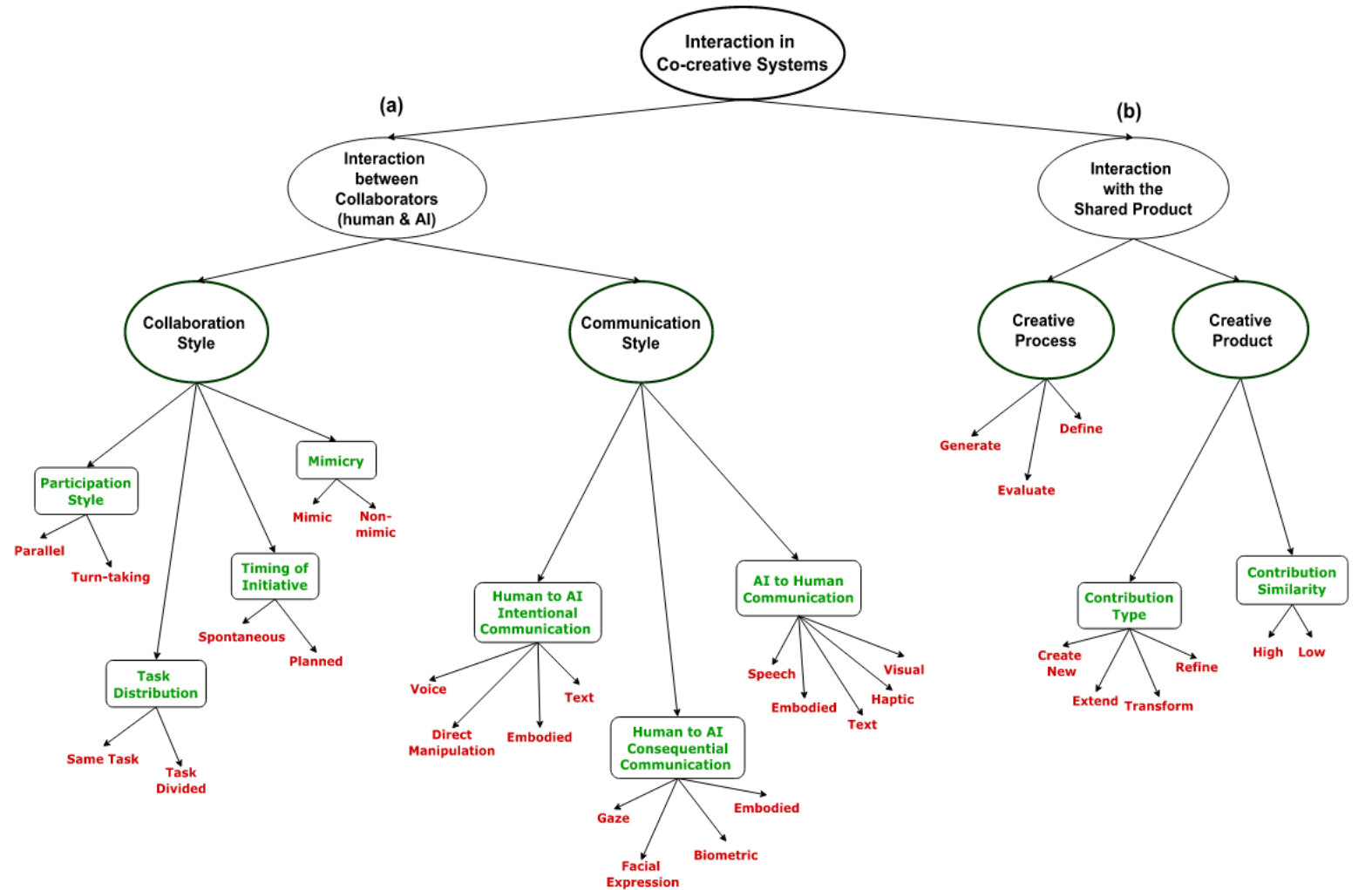
[Rezwana & Maher, 2022]

“**Interaction with the shared product**, in the context of a co-creative system, describes the ways in which the co-creator can **sense**, **contribute**, and **edit** the content of the emerging creative product.

Interaction between collaborators explains how the interaction between the co-creators is unfolding through time which includes **turn-taking**, timing of **initiative**, **communication** etc.

Participatory sense-making occurs [...] when both participants are **adapting their responses based on each other’s contribution** while maintaining an engaging interaction dynamic [...]

Structuring co-creation



Note: Beyond the conceptual framework itself (see image), this paper provides a comprehensive literature overview; it's a great starting point if you'd like to read more on this topic! 😊



Designing Creative AI Partners with COFI: A Framework for Modeling Interaction in Human-AI Co-Creative Systems

JEBA REZWANA and MARY LOU MAHER, University of North Carolina at Charlotte

67

Human-AI co-creativity involves both humans and AI collaborating on a shared creative product as partners. In a creative collaboration, interaction dynamics, such as turn-taking, contribution type, and communication, are the driving forces of the co-creative process. Therefore the interaction model is a critical and essential component for effective co-creative systems. There is relatively little research about interaction design in the co-creativity field, which is reflected in a lack of focus on interaction design in many existing co-creative systems. The primary focus of co-creativity research has been on the abilities of the AI. This article focuses on the importance of interaction design in co-creative systems with the development of the Co-Creative Framework for Interaction Design (COFI) that describes the broad scope of possibilities for interaction design in co-creative systems. Researchers can use COFI for modeling interaction in co-creative systems by exploring alternatives in this design space of interaction. COFI can also be beneficial while investigating and interpreting the interaction design of existing co-creative systems. We coded a dataset of existing 92 co-creative systems using COFI and analyzed the data to show how COFI provides a basis to categorize the interaction models of existing co-creative systems. We identify opportunities to shift the focus of interaction models in co-creativity to enable more communication between the user and AI leading to human-AI partnerships.

CCS Concepts: • Human-centered computing → Interaction design process and methods; Collaborative interaction;

Additional Key Words and Phrases: Human-AI co-creativity, co-creativity, interaction design, framework

ACM Reference format:
Jeba Rezwana and Mary Lou Maher. 2023. Designing Creative AI Partners with COFI: A Framework for Modeling Interaction in Human-AI Co-Creative Systems. *ACM Trans. Comput.-Hum. Interact.* 30, 5, Article 67 (September 2023), 28 pages.
<https://doi.org/10.1145/3519026>

1 INTRODUCTION

Computational creativity is an interdisciplinary field that applies artificial intelligence to develop computational systems capable of producing creative artifacts, ideas and performances [33]. Research in computational creativity has led to different types of creative systems that can be categorized based on their purposes: systems that generate novel and valuable creative products, systems that support human creativity, and systems that collaborate with the user on a shared creative product combining the creative ability of both the user and the AI [39]. Davis introduces

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1073-0516/2023/09-ART67 \$15.00
<https://doi.org/10.1145/3519026>

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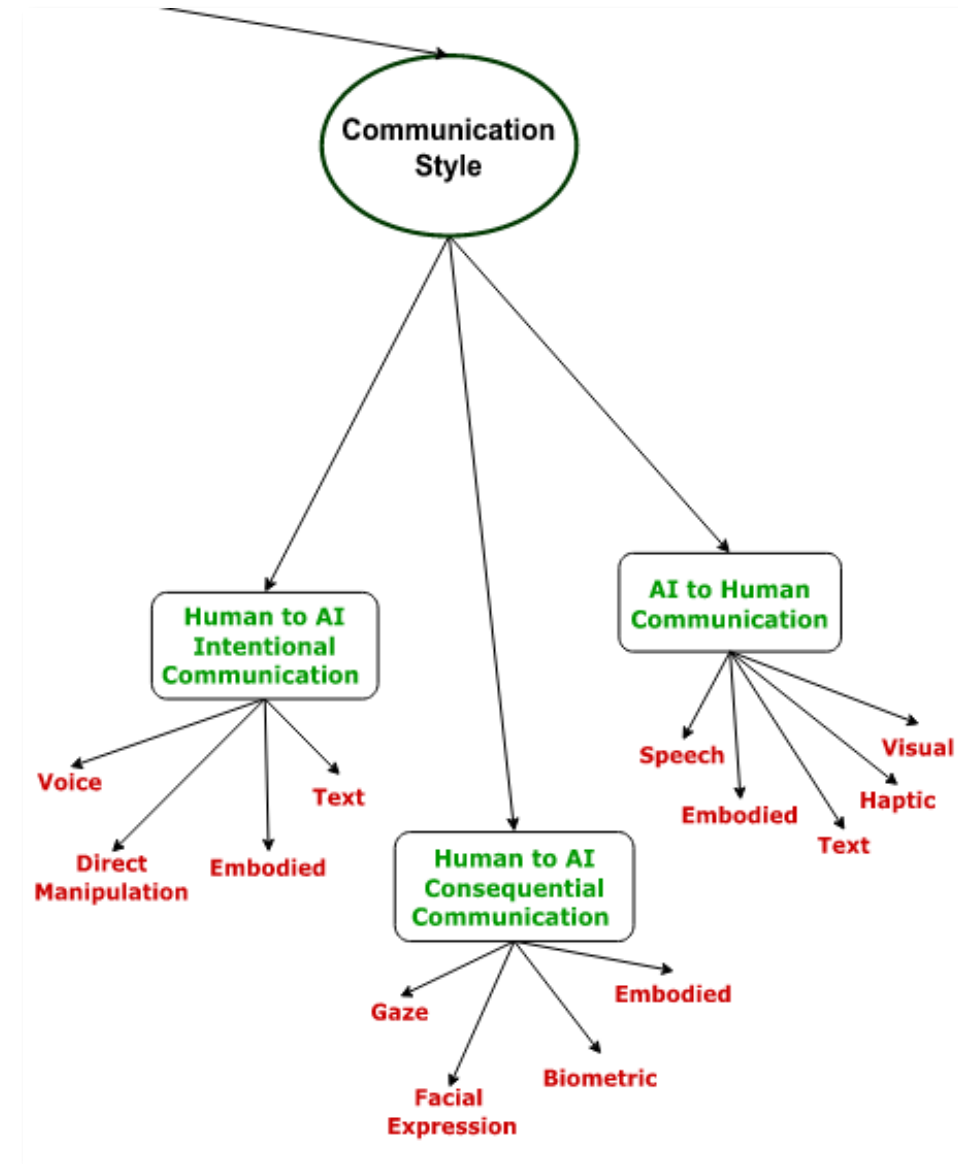
[Rezwana & Maher, 2022]

Structuring co-creation

Some “highlights” (e.g. for your projects & beyond)

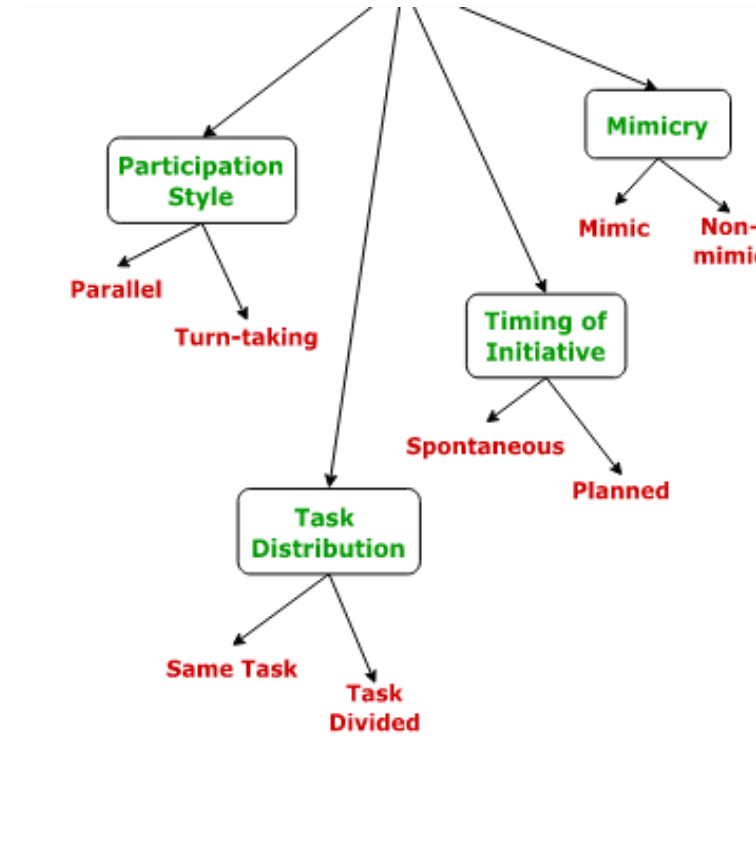
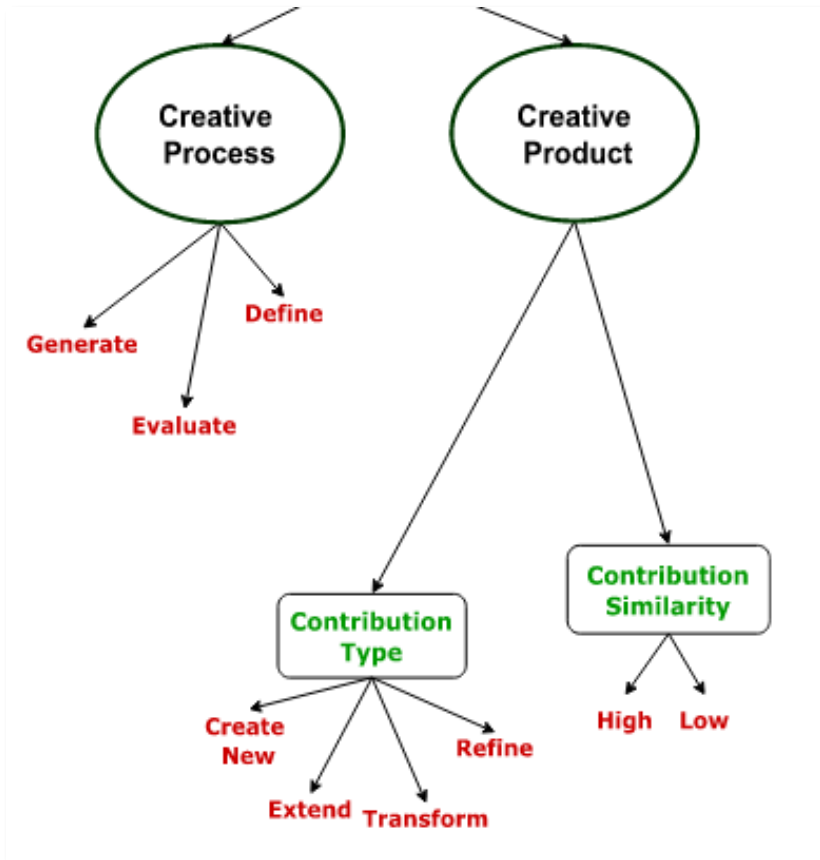
Thinking about these “communication styles” might be inspiring:

- **Intentional communication:**
What we usually think about for UIs (see prev. sessions)
- **Consequential communication:**
Signals from user to system that the user might not explicitly think about as interaction
→ highlights further input opportunities
- **AI-to-human communication:**
There’s more than screen-based output
→ highlights further output opportunities



Structuring co-creation

Some “highlights” (e.g. for your projects & beyond)



The framework captures key **concepts underlying the “roles”-terms** we’ve seen earlier. With these concepts, we can make the roles more specific & concrete than just giving them a name like “friend”.

What’s the **value** of this? E.g.:

- Analyse/compare UI designs
- Make your own design arguments concrete
(What are the options and why did we choose X?)

What's out there?

Trends in co-creative systems (in research), categorised using this framework

Their literature review identifies three “clusters” (the rows/colours):

Interaction between Collaborators							
Cluster No / Type of AI in Clusters	Collaboration Style				Communication Style		
	Participation Style	Task Distribution	Timing of Initiative	Mimicry	Human to AI Intentional Communication	Human to AI Consequential Communication	AI to human Communication
1 / Generative Pleasing Co-creative AI Agents (67 out of 92 systems)	Turn taking	Task Divided	Planned	Non-mimic	Direct Manipulation	None	None
2 / Improvisational Co-creative AI Agents (9 out of 92 systems)	Parallel	Single Task	Spontaneous	Mimic + Non-mimic	None	None	None
3 / Advisory Co-creative AI Agents (16 out of 92 systems)	Turn taking	Task Divided	Planned	Non-mimic	Direct Manipulation	None	None

Interaction with the Shared Product			
Cluster No / Type of AI in Clusters	Creative Product		Creative Process
	Contribution Type	Contribution Similarity	
1 / Generative Pleasing Co-creative AI Agents (67 out of 92 systems)	Create New	High	Generation
2 / Improvisational Co-creative AI Agents (9 out of 92 systems)	Create New	High + Low	Generation
3 / Advisory Co-creative AI Agents (16 out of 92 systems)	Refine	High + Low	Generation + Evaluation

“This [cluster 1] shows that **most of the co-creative systems in the dataset utilize turn-taking** as the participation style. Therefore, each of the **collaborators must wait until the partner finishes their turn**. This interaction model uses **“planned” timing of initiative** which is an indication of non-improvisational co-creativity.”

Example

“Duet Draw”



[Oh et al., 2018]

Example: More on initiative & turn taking

In “Duet Draw”

“There are **two initiative styles**: **Lead** and **Assist**.

In the **Lead** style, users complete their pictures with the help of the AI. In this mode, **users take the initiative**. Users draw a major portion of the figure, and the AI then carries out secondary tasks.

In contrast, in the **Assist** style, users help AI to complete the picture. In this mode, the **AI takes the initiative**. The AI draws the main parts of the picture and asks users to complete supplementary / subsidiary parts.”

[Oh et al., 2018]

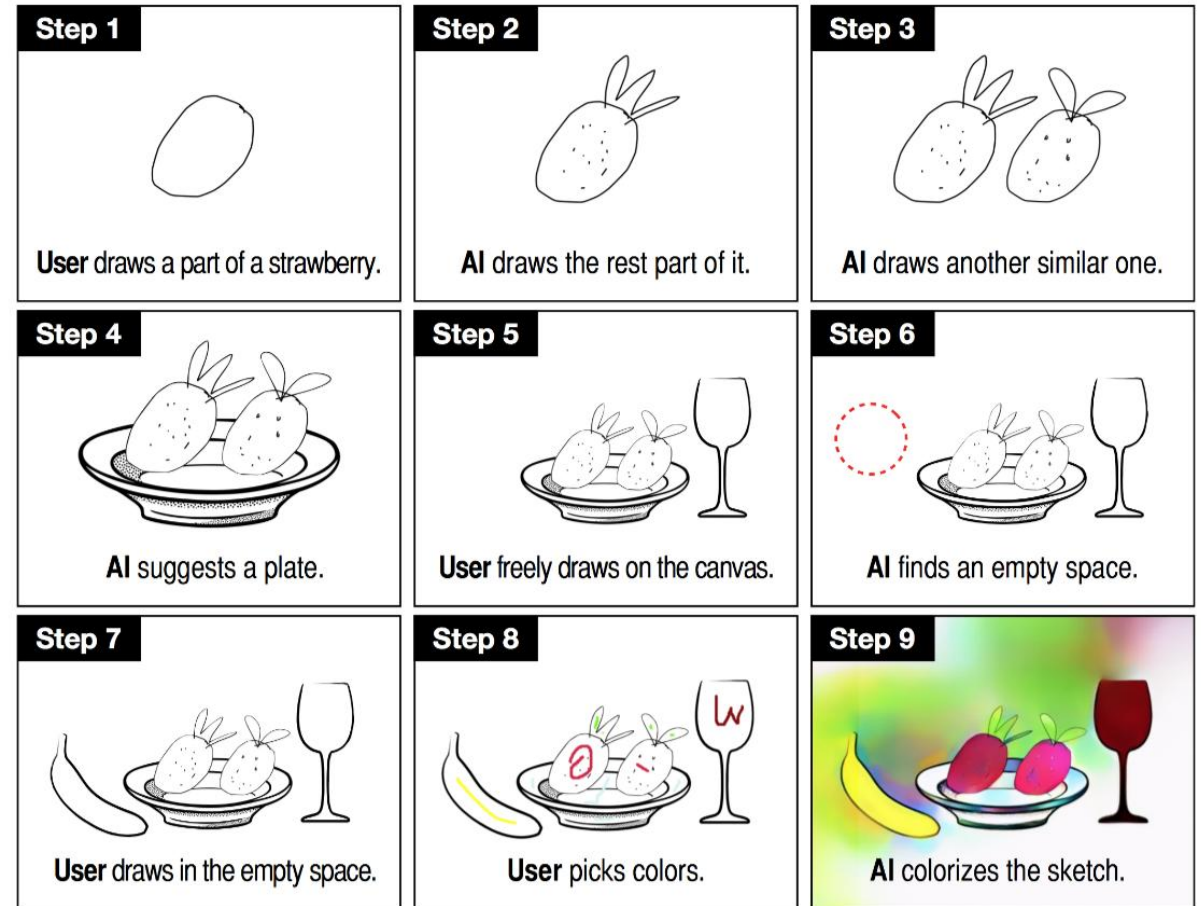


Figure 1. Drawing using DuetDraw in the Lead mode. With DuetDraw, users and AI can collaboratively draw pictures.

Example: More on communication

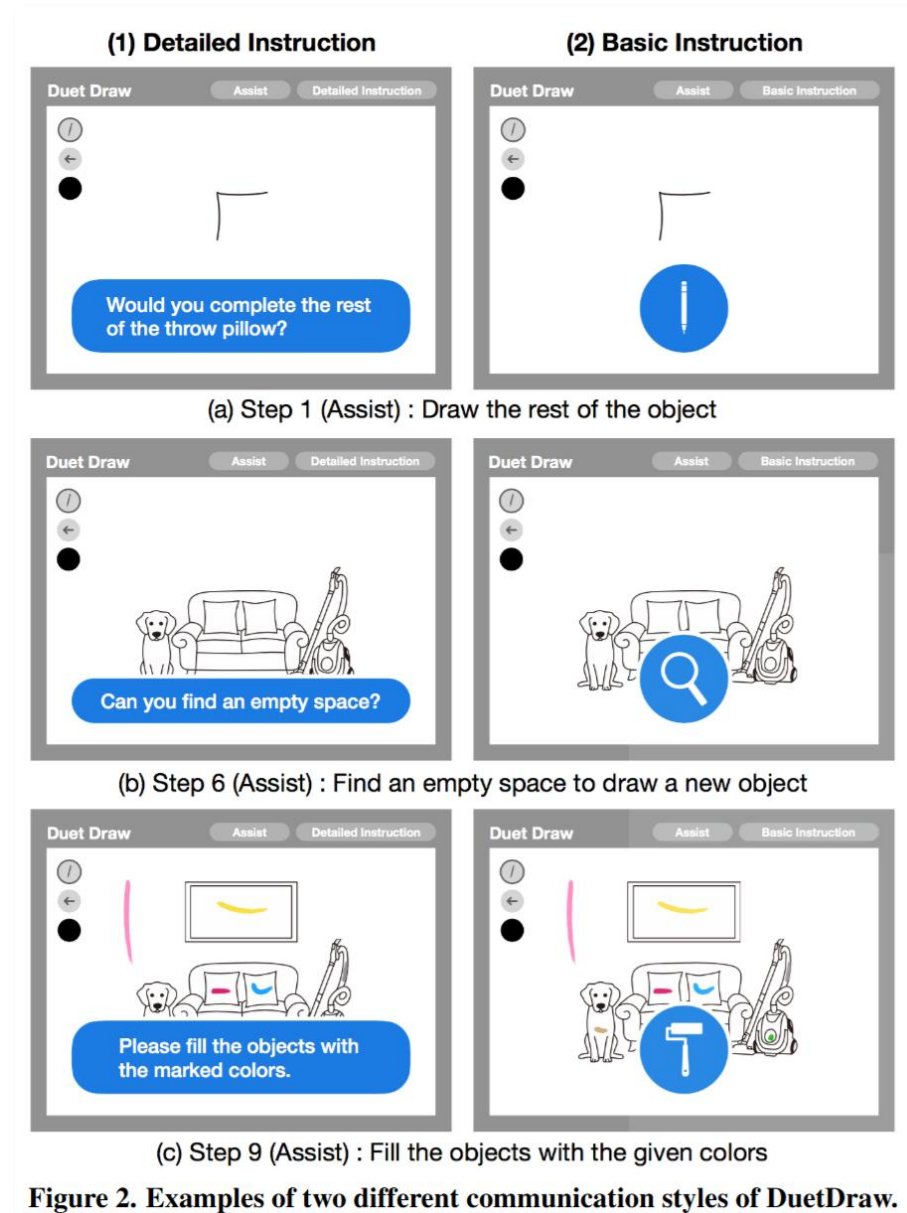
In “Duet Draw”

“There are **two styles of communication**: **Detailed Instruction** and **Basic Instruction**.”

In **Detailed Instruction**, the AI explains each step and guides the user. At the bottom of the interface, an **instruction is displayed as a message**, and users can confirm the message by tapping yes or no buttons.

On the contrary, in **Basic Instruction**, the AI automatically proceeds to the next step with basic notifications. An **instruction is displayed as an icon** on the canvas”

[Oh et al., 2018]



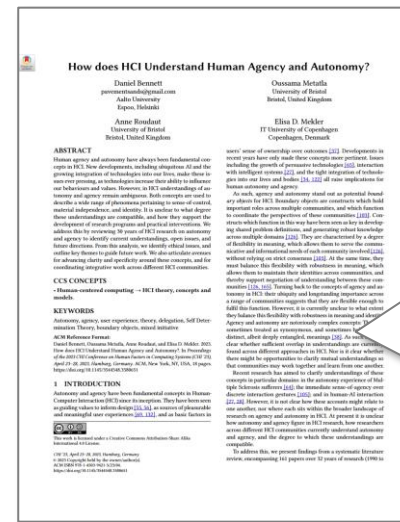
Further aspects

Recommended reading

■ Agency and autonomy

Various understandings – how to gain clarity?

See: “How does HCI Understand Human Agency and Autonomy?” ([link](#))



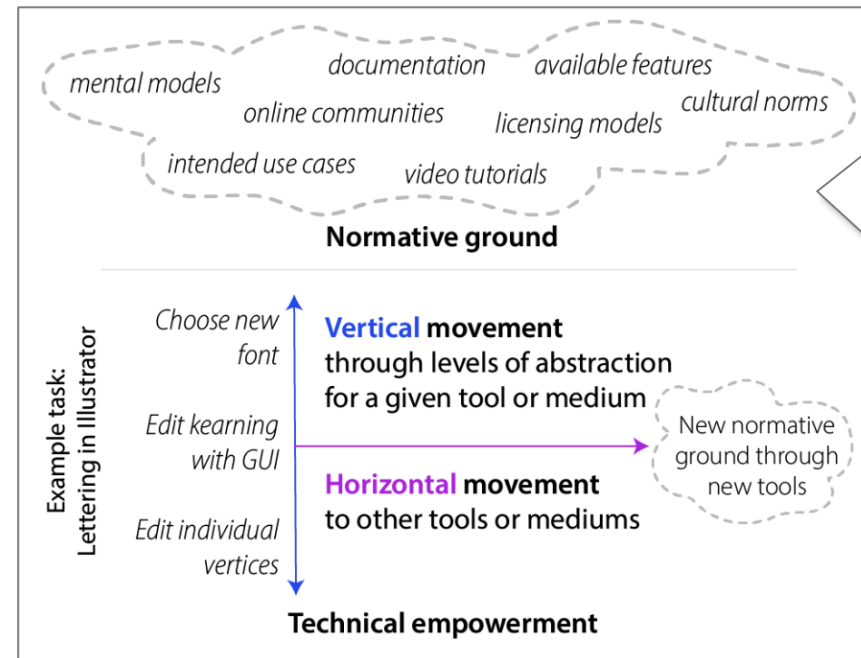
“Our analysis identifies both the high value of these concepts for HCI research, their flexibility, and the degree of ambiguity in how they are understood. [...] four aspects which help clarify understandings of agency and autonomy in HCI:

1. issues of self-causality and personal identity;
2. the experience and material expression of autonomy and agency;
3. particular time-scales;
4. emphasis on independence and interdependence.”

■ Empowerment in creativity support tools

Do (co-creative) systems give users “power-to” create, or do they give designers & developers “power-over” what users can create?

See: “Beyond the Artifact: Power as a Lens for Creativity Support Tools” ([link](#))



“Top: A tool’s normative ground incorporates many factors that shape how creative practitioners think about and use the tool, such as features, communities, and cultural norms.

Bottom: We argue vertical movement through levels of abstraction and horizontal movement to other tools are crucial for technically empowering creative practitioners.”

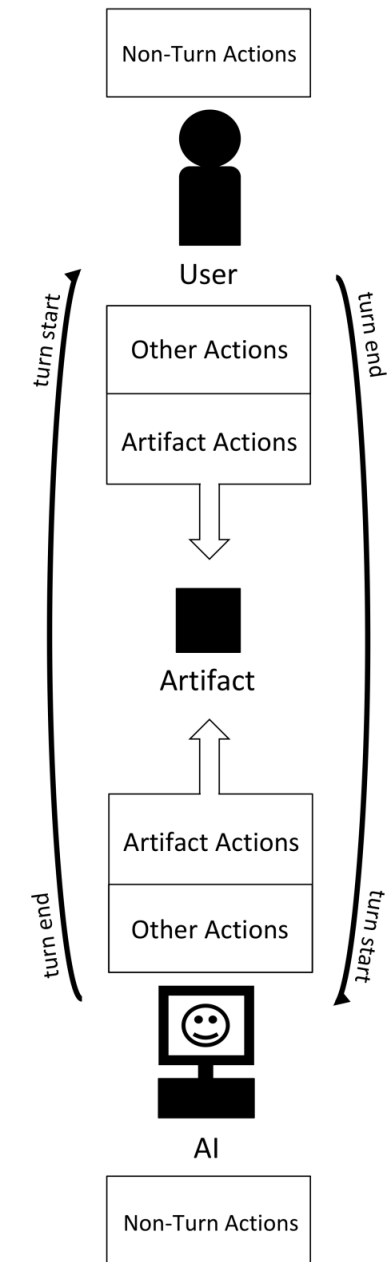
Analysing co-creative interaction

By structuring interaction in “turns”
and different types of “actions”
that user and/or system can take
when it’s (not) their turn

Framework for analysing co-creative AI

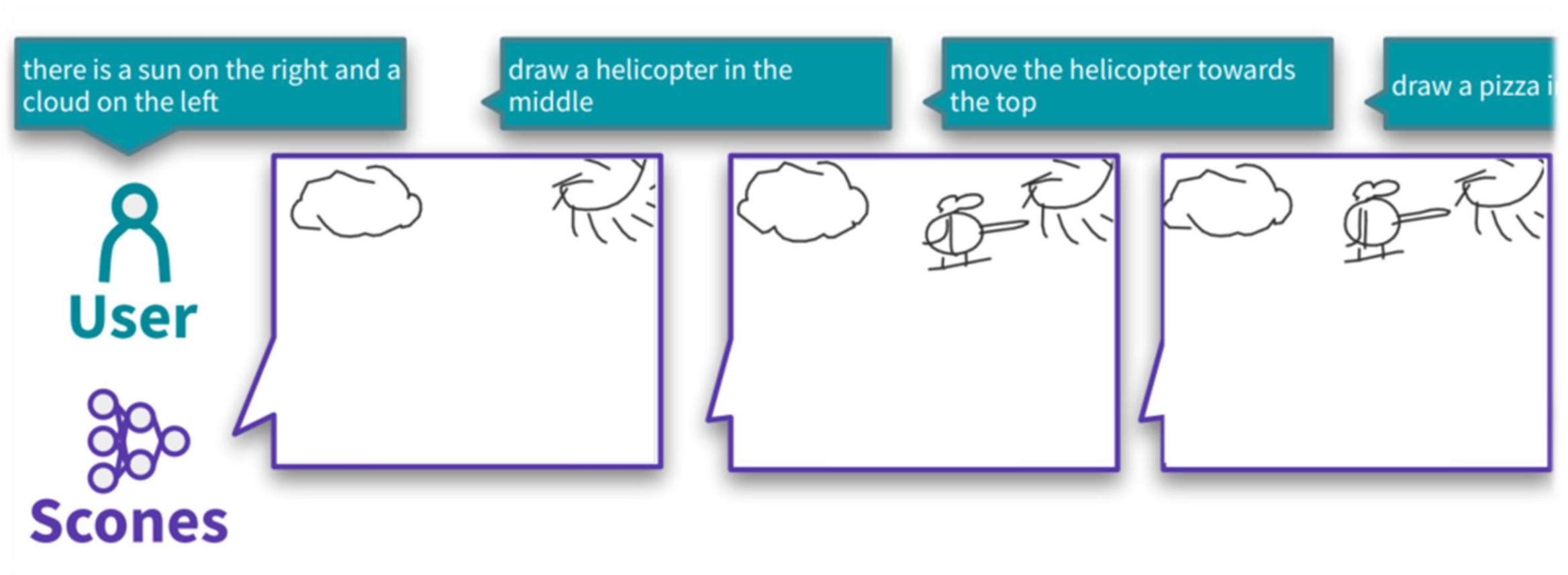
By Guzdial & Riedl, 2019

- **Start:** How does the interaction start? Who goes first?
- **Actions:** What actions is each partner capable of
 - **Artifact Actions:** How can each partner impact the artifact?
 - **Other Actions:** Actions that don't change the artifact?
E.g. explanations
 - **Non-Turn Actions:** Actions outside of their turn?
E.g. AI learning from the user's actions; or user exploring the AI model
- **Turns:** How do turns begin/end? Who decides when?
- **End:** How does the interaction end?



Example: “Scones”

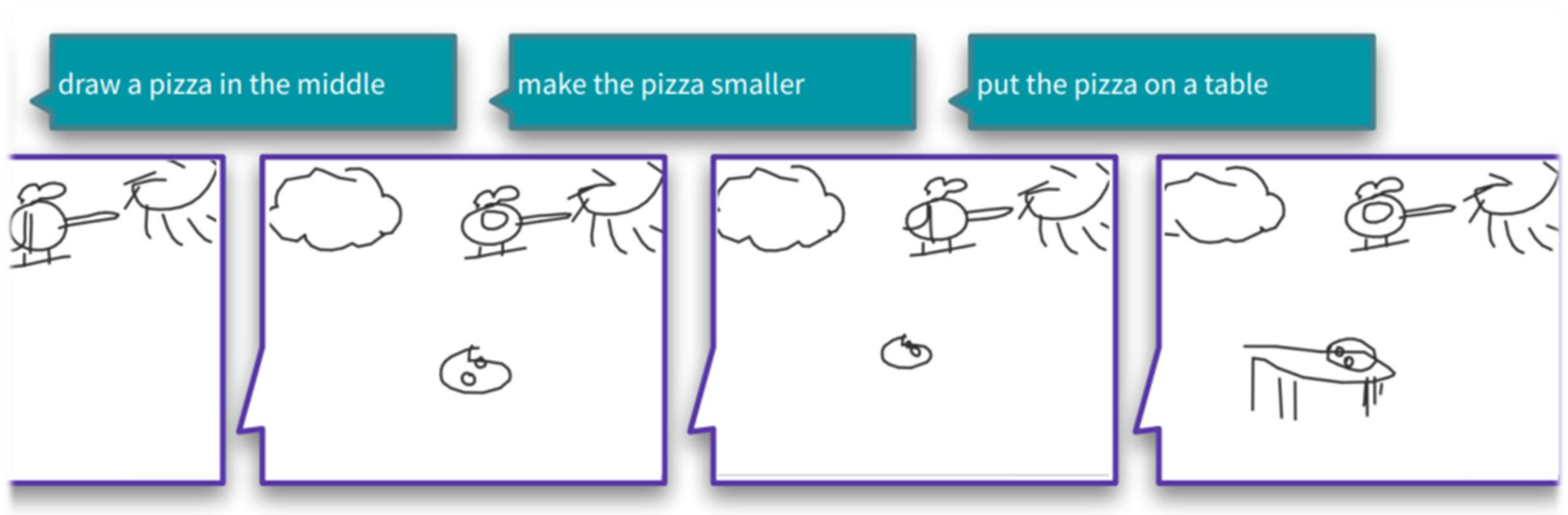
Applying the framework to analyse a co-creative system



[Huang et al., 2020]

Example: “Scones”

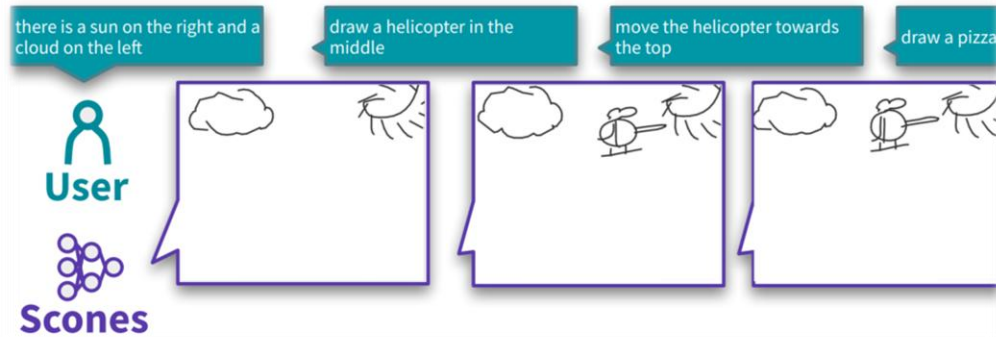
Applying the framework to analyse a co-creative system



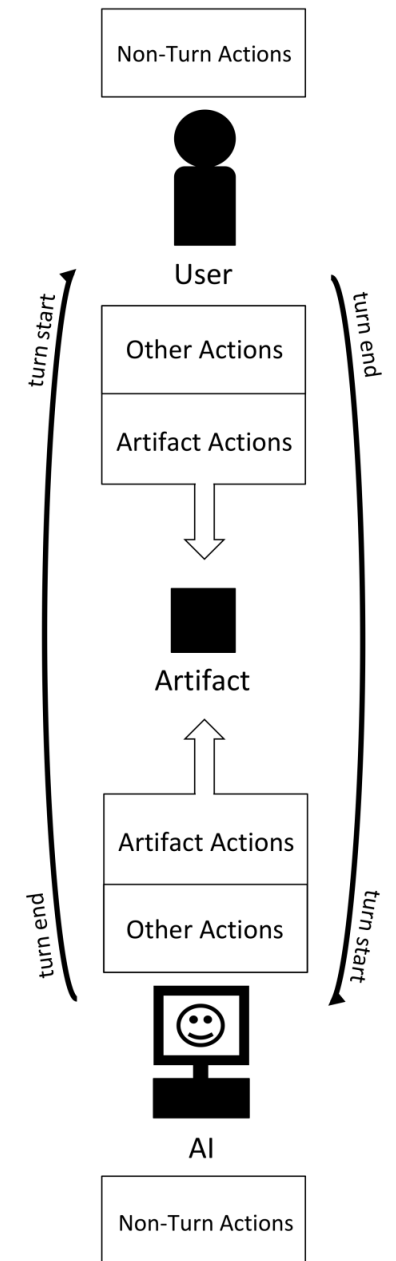
[Huang et al., 2020]

Activity

Applying the framework

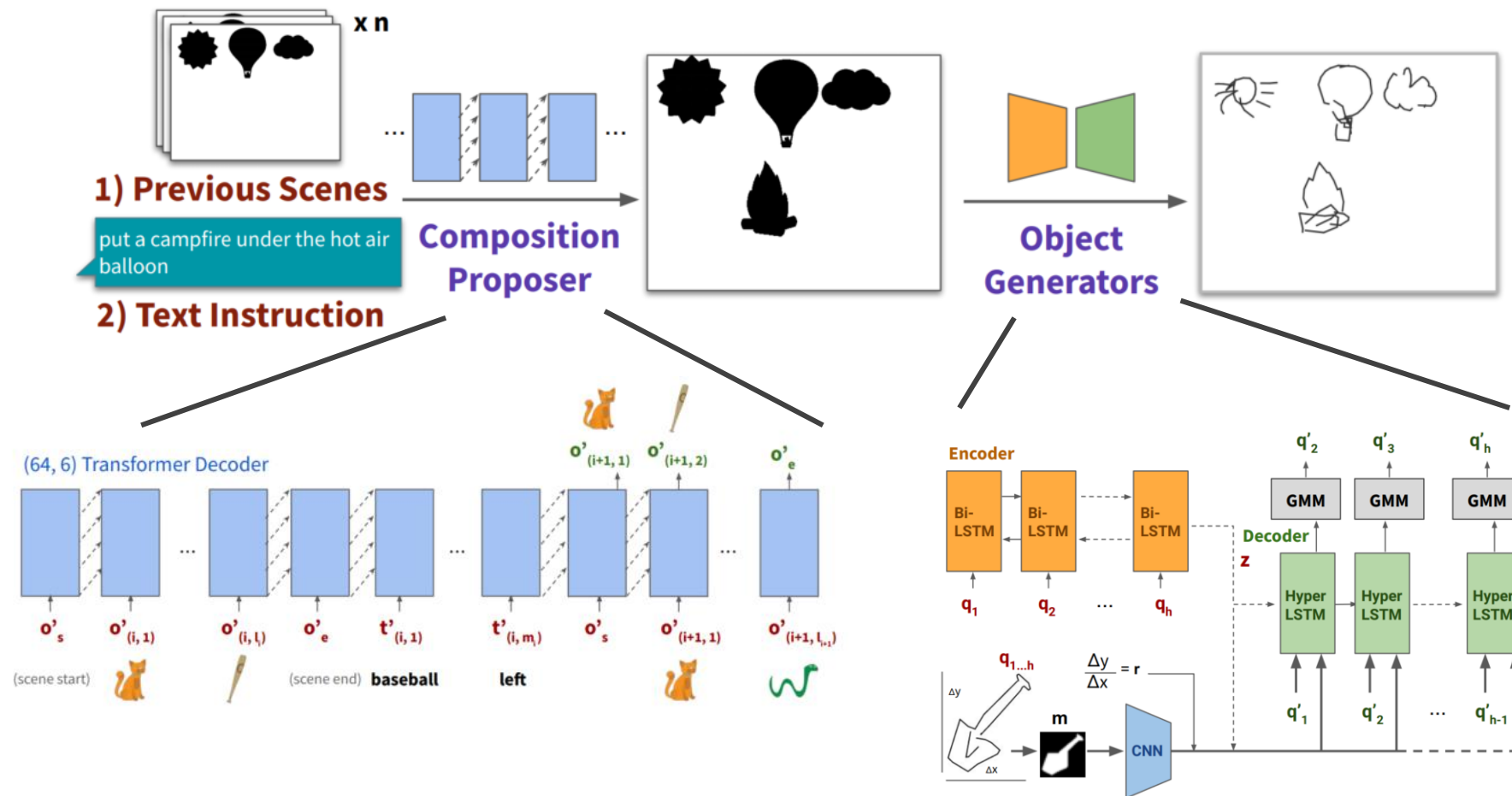


Aspect	User	System
Start		
Artifact actions		
Other actions		
Non-turn actions		
Turns		
End		



Example: “Scones”

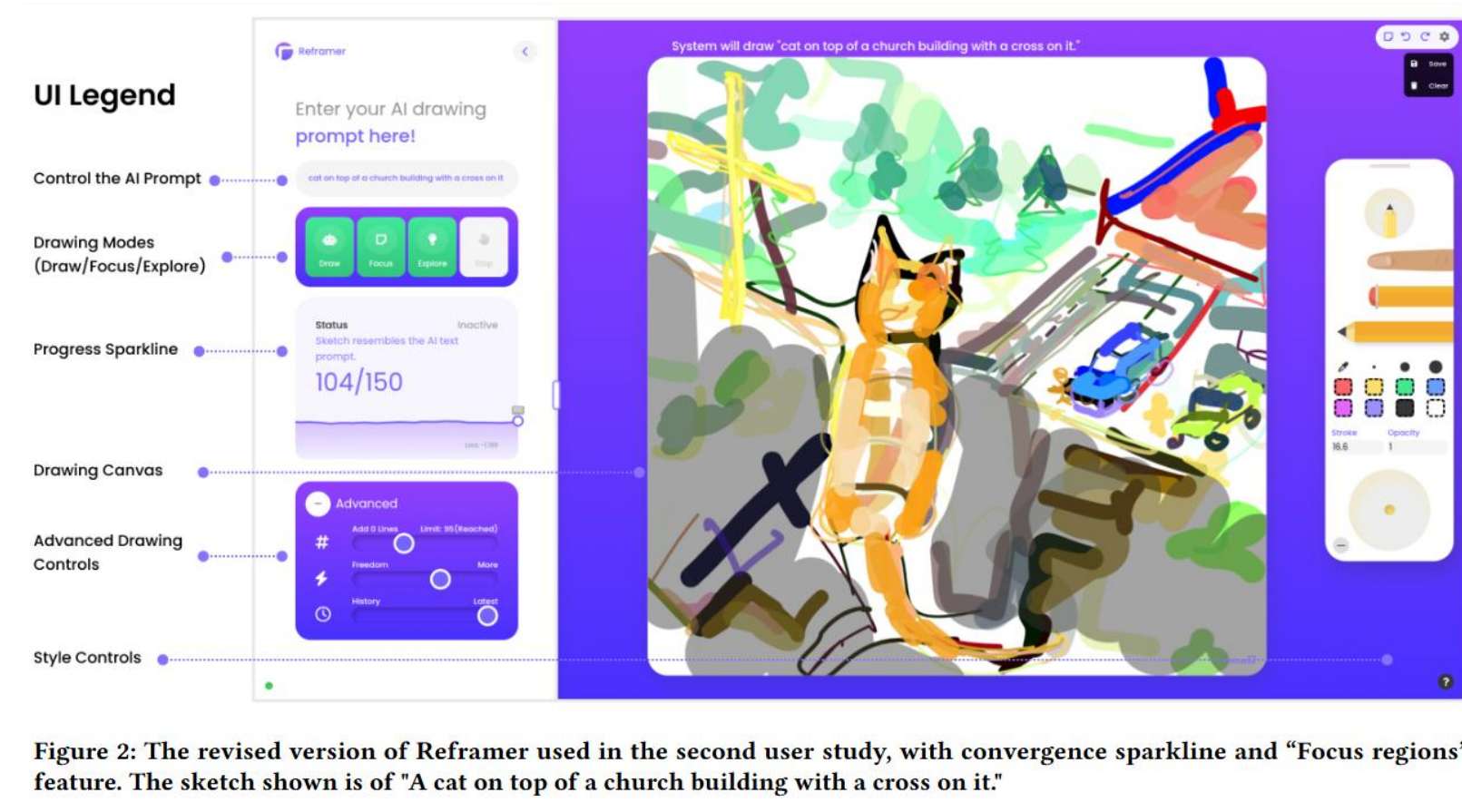
Implementation overview (just if you're curious, not relevant for the application of the framework)



[Huang et al., 2020]

Beyond “turns”

Not all co-creative interaction designs fit “turn-taking”, e.g. here’s a study with a parallel sketching agent:



“Reframer facilitates fluid, **real-time shared drawing experiences with a generative AI**, and our studies indicated this novel interaction paradigm enables a **bi-directional exchange of ideas** between its users and the AI system.

This mutual influence can create moments of serendipitous discovery and design reformulation, which both delighted our users and contributed to their overall positive experiences. We **contrasted** these engaging and positive “**agentive flow**” experiences with traditional notions of creativity support that focus on maximally **unobtrusive tools**.

Through our thematic analysis, we established **users move between agentive and non-agentive creative flow states** as they adapt to and are disrupted by a tool’s functions, with different features helping at some points in the process and hindering at others.”

[Lawton et al., 2020]

Video: <https://www.youtube.com/watch?v=2AfUK4skDZk>



"A family of peaches having a picnic on a hill"

Pitfalls

What could go wrong?



Pitfalls

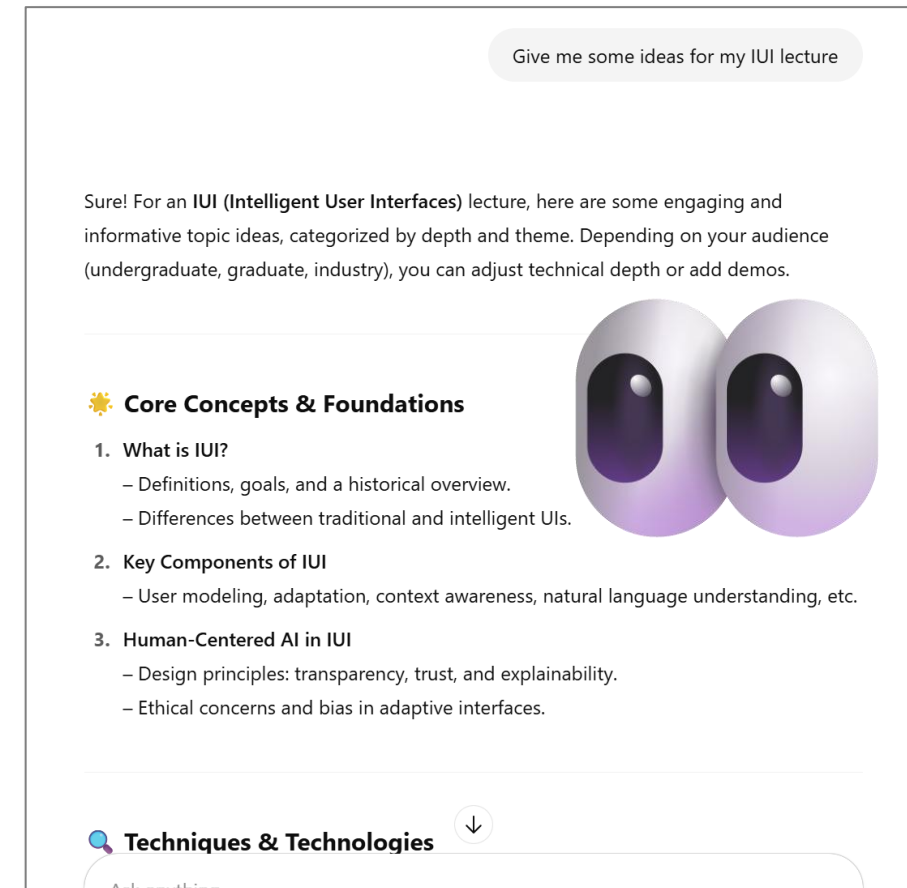
Creativity support vs “fixations”?

Interaction with **generative systems** can create “fixations” on AI output

- E.g. writing with suggestions may **influence users’ text (and opinions)** ([link](#))
- E.g. ideation with image generation may **reduce number & variety of ideas** ([link](#))

Maybe you have experiences with this already?

- E.g. reading a chatbot response and thinking “yeah, sounds ok”
 - But was it what you would have created from scratch?
 - Or: Could you re-prompt without being anchored to the previous output?
- E.g. in our creature example:
 - Once you’ve “peeked”, there’s no going back to an “independent state” of thinking!



Q: Using the concepts from this session, can you think of design choices (or “AI roles”) that you expect would help reduce unwanted influences on the user’s thinking/creativity?

More potential pitfalls

Name	Affected aspects	Problem description	Example	How it might have happened (examples)	How it might be addressed (examples)
Limited AI					
1 Invisible AI boundaries	model, creativity, exploration	A (generative) AI component imposes unknown restrictions on creativity and exploration.	An AI face image editor cannot make faces bald without also turning them male-looking.	Model with limited generalisability beyond training data, and entangled or nonsensical (latent) dimensions w.r.t. human understanding.	UI: Show boundaries e.g. via uncertainty, samples, precision/recall [21]. AI: Improve generalisability, disentanglement; consider narrowing scope.
2 Lack of expressive interaction	usability, creativity, exploration	The UI imposes a "bottleneck" on creative use of the AI.	Image generator is controlled with many 1D inputs for a high D latent space [22] - vs. rich image editor tools like brushes.	Fine-grained AI control is difficult. "Conservative" UI design focused on ensuring input stays in (training) data distribution.	Human-centred design with target group, e.g. to inform preferable tradeoffs of UI expressiveness and model "breaking points".
3 False sense of proficiency	trust, reliability	AI suggests answers or completions that the user cannot verify or that generate a false sense of proficiency.	When prompted to complete a sentence about the population of a large city the AI delivers a reasonable number that could be correct - but might not be.	Language models are capable of memorizing excerpts of text and reproducing them when prompted with a similar context.	Learn an additional model, that can attribute generated content to an explicit source to allow for verifying correctness.
Too much AI					
4 Conflicts of territory	usability, UX, control	AI overwrites what the user has manually created/edited.	In a co-creative text editor, the user replaces terms in generated text. Later, the AI (partly) reverts these changes.	Language model optimised for word probability and user's term was less likely.	Keep track of user edits to protect them, ask for confirmation before changes, or to integrate this info into inference.
5 Agony of choice	usability, UX, productivity	AI provides overwhelming amount/detail of content that distracts or creates agony of choice.	An AI photo editor displays an excessive number of suggested variants. The resulting small previews make it hard to discern and decide.	UI design process was focused on showing AI capabilities instead of user needs.	Clarifying use cases and support needs, responsive / malleable UI concepts, changeable user settings.
6 Time waster	usability, UX, productivity	AI interrupts user or draws attention away from the creative task itself.	A co-creative music composition tool continuously shows melody completions, which keep the user busy with exploring or understanding the system instead of their ideas.	Same as above. Also: Timing of the AI's involvement not tested with users or varying preferences between users.	Same as above. Attention-aware UI (e.g. AI waits to not disrupt user's focused work; or stops suggestions if user has explored it for a while).
Beyond use					
7 AI bias	accountability, fairness, transparency	AI suggestions are biased in a certain unwanted way, w.r.t. human meaning and values.	An AI story generator writes gender-stereotypical protagonists (e.g. w.r.t. roles/occupations).	AI picked up biases in the training data or created bias through its learning method. Development process unaware of biases.	Design for easy human revision/rejection. Addressing AI bias (e.g. see [23, 24]). Learning from user feedback/actions.
8 Conflict of Creation & Responsibility	creativity, responsibility, ownership	A system and a user collaborate to create an output. Ownership and responsibility are unclear.	In a co-creative text editor the AI suggests formulations that appear verbatim in the training data. Who is the owner of the resulting text?	Co-creative systems operate on a continuum between user and system creation, challenging attributions of ownership.	Should we attribute an AI and training data providers as contributors? Do we need systems to check for (accidental) plagiarism?
9 User and Data Privacy	privacy, responsibility	Private data may be exposed through the AI system or its training data.	1) A user A works with a cloud-based AI text creator and their data is transmitted unencrypted. 2) The AI reveals (private parts of) another user B's data to A (e.g. [25, 26]).	AI models are trained on a large corpus of data and can sometimes default to replicating this data when prompted.	Remove private information from training sets and work with AI either encrypted or locally.

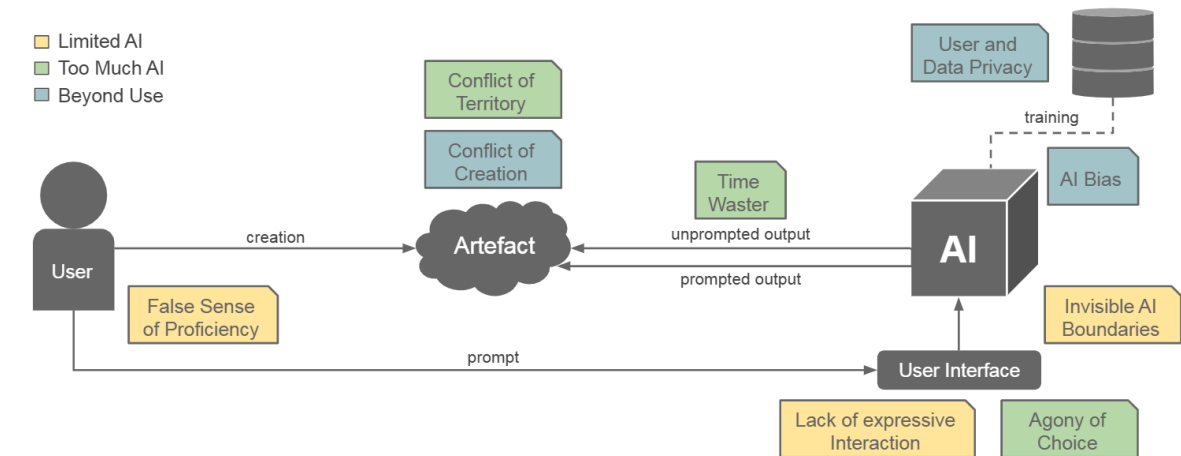


Figure 1: Visualisation of our underlying mental model of the interaction loop in human-AI co-creative systems. We place our identified pitfalls (see Table 1) in this loop based on the position where they most likely occur.

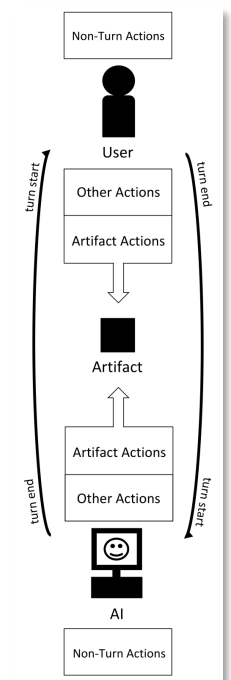
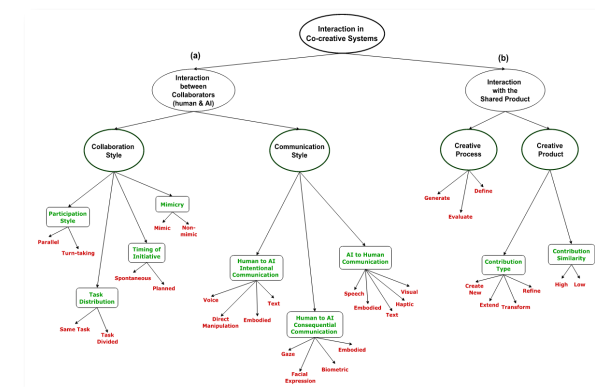
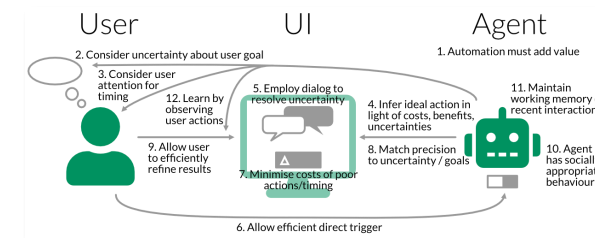
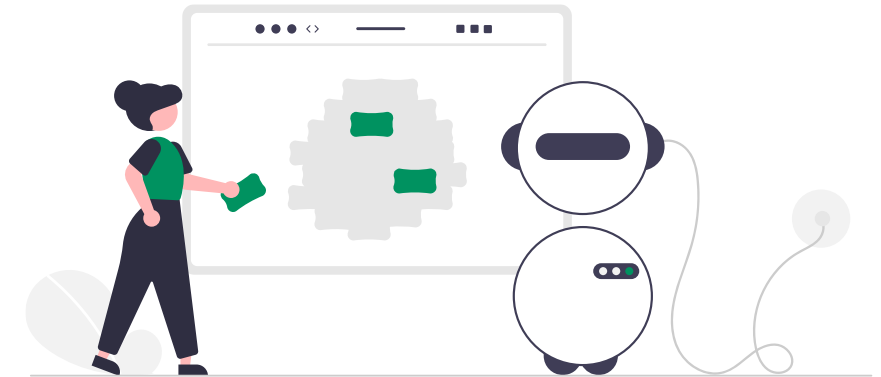
Note: We brainstormed these potential pitfalls in our research group here at UBT for a workshop in 2021. If you're interested and read it and have some further ideas for pitfalls (or recent examples), please let us know!

[Buschek et al., 2021]

Conclusion

Summary & conclusion

- In **co-creative systems**, both **user and system contribute** (steps) towards creating a digital artefact
- This often motivates using “generative AI”; but beyond LLMs, image generation, etc. this could be any **model that predicts next creative actions** given the previous actions and/or current state of the artefact
- We’ve covered three selected **conceptual frameworks to help us think systematically** about UIs and interaction design for generative systems and co-creation. This gives us three lenses:
 - **Analytical:** Describe a UI/interaction design with these concepts & terms
 - **Constructive:** Use the concepts to systematically consider your options. E.g. to go beyond the currently dominant “system-as-a-content-generator”.
 - **Critical:** Ask questions such as:
Are some design choices better than others? For a use-case & target group?
To avoid pitfalls, such as unwanted impacts on user and artefact?”



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